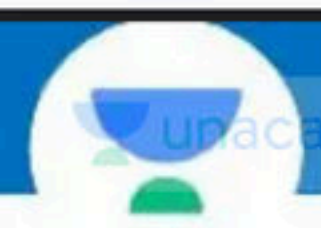




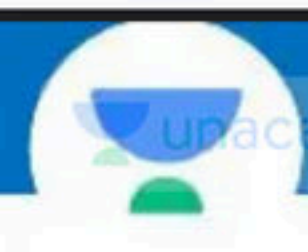
# Unit - 5 (Human Physiology) - 200 MCQs

NCERT Nichod : Biology Practice Series NEET 2025



# 200 MCQs on Human Physiology

⇒  
Good evening  
↓  
audible?  
↓  
Ready?

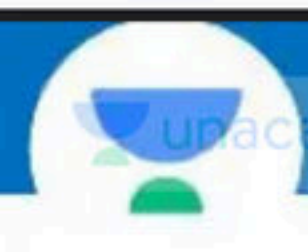


## PART-1 (SECTION-A)

V.V.I.P  
1. Which is a correct statement :-

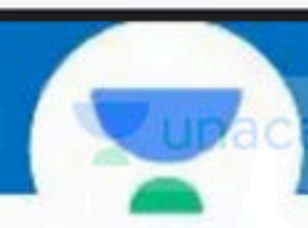
- (a) Hypertension is caused by anaemia ✗
  - (b) Cardiac output depends on stroke volume ✓
  - (c) Blood capillary is single walled ✓
  - (d) Man has hepatic portal system ✓
- (1) Only a & b
- (2) Only b & c
- ✓ (3) Only b, c & d
- (4) Only a, b & c





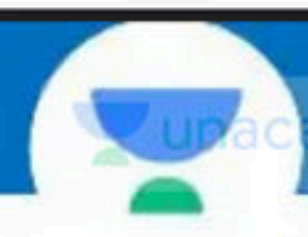
**2. Which one is Rh incompatible marriage :-**

- ✓ (1) Rh negative female and Rh positive male
- (2) Rh positive female and Rh positive male
- (3) Rh negative female and Rh negative male
- (4) Rh positive female and Rh negative male



③ Which neural signals decrease the rate of heart beat:-

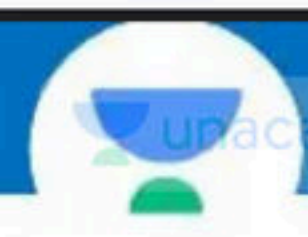
- (1) Neural signals through the sympathetic nerves
- ✓ (2) Neural signals through the parasympathetic nerves
- (3) Hormone of adrenal medulla
- (4) Both 1 and 3



↓  
**4) Which statement is incorrect :**

- (1) Neutrophils are the most abundant cells about 60-65 percent of the total WBCs and basophils are the least about 0.5-1 percent among them
- (2) The T-wave represents the return of the ventricles from excited to normal state
- ✓ (3) The P-wave represents ~~repolarisation~~ <sup>depolarisation</sup> of the atria
- (4) A healthy individual has 12-16 gms of haemoglobin in every 100 ml of blood





⑤ Which is a correct statement :-

(a) Fibrinogen helps in maintenance of ~~osmotic pressure~~ <sup>Blood clotting</sup> X

(b) Glucose is absent in plasma X

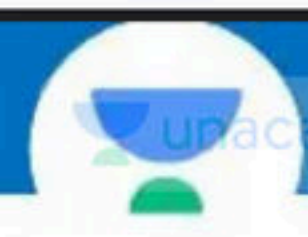
✓(c) Serum does not clot

✓(d) Lymph is a connective tissue

(1) a & b    (2) b & c

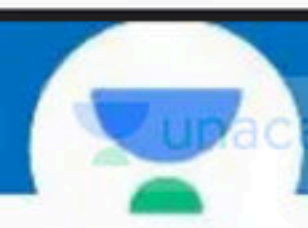
✓(3) c & d    (4) a & d





6. Which disorder is associated with Rh antigens :

- (1) Hepatitis
- (2) Anaemia ✓ ↩
- (3) Erythroblastosis foetalis ✓
- ✓ (4) Both 2 and 3



7) Which is a correct statement : -  $\rightarrow SA$

(a) AV node is called as pace maker  $\times$

(b) Atrial systole causes ~~70%~~ 30% ventricular filling  $\times$

$\checkmark$  (c) Cardiac output depends on ventricular filling  $\checkmark$

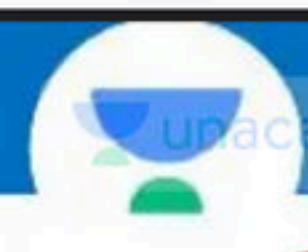
(d) Blood pumped during a cardiac cycle is ~~cardiac output~~  
stroke volume

(1) Only c & d

$\checkmark$  (2) Only c

(3) Only d

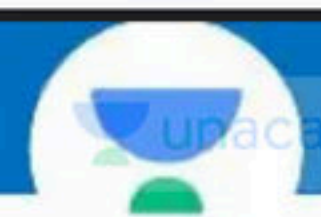
(4) Only a & d



⑧ What is correct for blood group 'O' :

- ✓ (1) No antigens but both anti-A and anti-B antibodies are present
- (2) A antigen and both anti-A and anti-B antibody are present
- (3) Antigen and antibody both absent
- (4) A and B antigens and anti-A, anti-B antibodies are present





9) Which is a correct matching :-

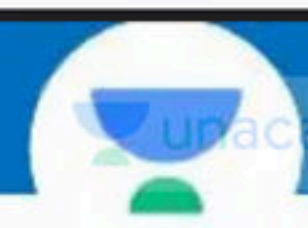
Column-I

Column-II

- |                     |                                   |
|---------------------|-----------------------------------|
| (a) Lymph capillary | i. Artificial pacemaker           |
| (b) Vein            | ii. Wall of artery                |
| (c) Tunica media    | iii. S.V.C. ( superior vena cava) |
| (d) Cardiac arrest  | iv. Lacteal                       |
- 

- ✓ (1) a-iv, b-iii, c-ii, d-i  
(2) a-i, b-iii, c-ii, d-iv  
(3) a-iii, b-ii, c-i, d-iv  
(4) a-i, b-ii, c-iii, d-iv





**10** Given below are two statements :

**Statement I :**

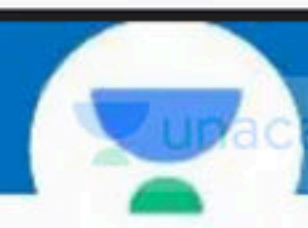
In adult man, normal BP is 120/80 mm Hg

**Statement-II :**

Chordae tendinae are present in ventricles

**Choose the correct answer from the option given below:**

- (1) Both Statement I and Statement II are incorrect
- (2) Statement I is correct but Statement II is incorrect
- (3) Statement I is incorrect but Statement II is correct
- ✓ (4) Both Statement I and Statement II are correct



11. Given below are two statements :

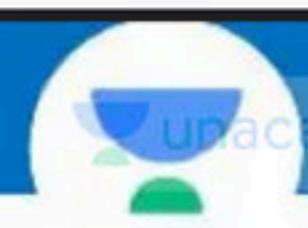
Statement I :

One cardiac cycle is completed in 0.8 sec. ✓

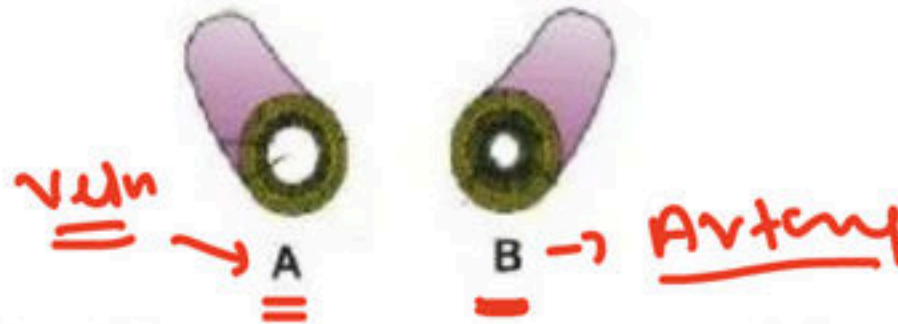
Statement-II :

Arterial depolarization is represented by <sup>P-wave</sup> ~~T-wave~~ ✗ Choose the correct answer from the option given below:

- (1) Both Statement I and Statement II are incorrect
- ✓(2) Statement I is correct but Statement II is incorrect
- (3) Statement I is incorrect but Statement II is correct
- (4) Both Statement I and Statement II are correct

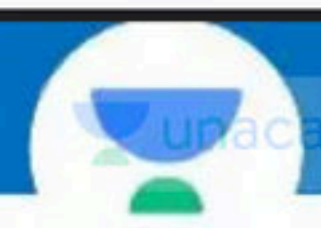


12. Choose the correct option according to given figure.



- (1) A – Artery : Tunica media is thin as compare to vein.
- (2) A – Vein : Tunica media is thick as compare to artery.
- ✓ (3) B – Artery : Tunica media is thick as compare to vein.
- (4) B – Vein : Tunica media is thick as compare to artery.





13 Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R)

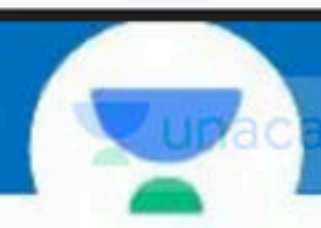
✓ Assertion (A) : Human heart is myogenic

✓ Reason (R) : The nodal musculature has the ability to generate action potential without any external stimuli so it is auto excitable and is responsible for maintaining the rhythmic contractile activity.

choose the correct answer from the options given below :

- (1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- (2) (A) is correct but (R) is not correct
- (3) (A) is not correct but (R) is correct
- ✓ (4) Both (A) and (R) are correct and (R) is the correct explanation of (A)





14. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R)

✓ Assertion (A) : Heart failure is not the same as cardiac arrest or a heart attack.

✗ Reason (R) : In heart failure means heart muscles is suddenly damaged by an indadequate blood supply. ✗

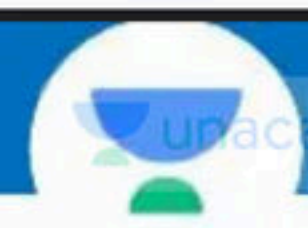
choose the correct answer from the options given below :

(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

✓ (2) (A) is correct but (R) is not correct

(3) (A) is not correct but (R) is correct

(4) Both (A) and (R) are correct and (R) is the correct explanation of (A)



15. Given below are two statements :

Statement I : *thick*

Arteries are ~~thin~~-walled and have narrow lumen as compared to veins

Statement-II :

*heart*  
Angina is acute chest pain when the blood circulation to the brain is reduced

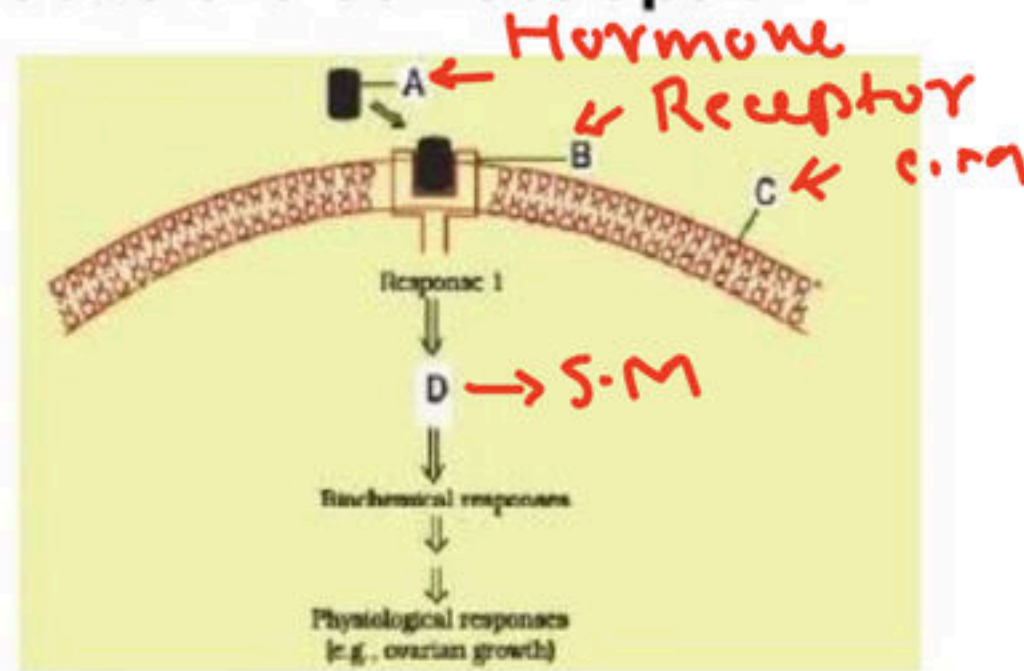
Choose the correct answer from the option given below:

- ✓(1) Both Statement I and Statement II are incorrect
- (2) Statement I is correct but Statement II is incorrect
- (3) Statement I is incorrect but Statement II is correct
- (4) Both Statement I and Statement II are correct



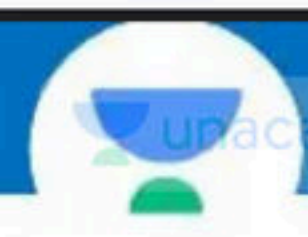


16. Identify A to D and choose the correct option :



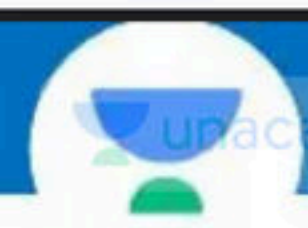
- ✓(1) A - Hormone, B - receptor, C - Cell membrane, D -Secondary messenger
- (2) A - Hormone, B - receptor, C - Cell membrane, D -Primary messenger
- (3) A - Receptor, B - Hormone, C - Cell membrane, D -primary messenger
- (4) A - Receptor, B - Hormone, C - Cell membrane, D -Secondary messenger





**17. Melatonin is a hormone secreted by :**

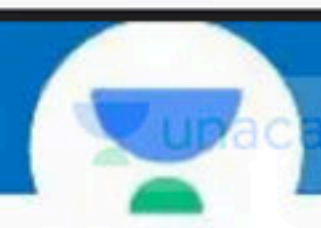
- (1) Pituitary gland
- ✓ (2) Pineal gland
- (3) Thymus gland
- (4) Adrenal cortex



18. The blood calcium level is lowered by the deficiency of :

- (1) Thyroxine
- (2) Calcitonin
- ✓(3) Parathormone
- (4) Both calcitonin and parathormone

→ B.  $Ca^{+2}$  ↑



19. Which of the following hormones are released by anterior lobe of pituitary :

✓ I. GH

✓ II. PRL

✓ III. TSH

✓ IV. ACTH

✓ V. LH

VI. Oxytocin ✗

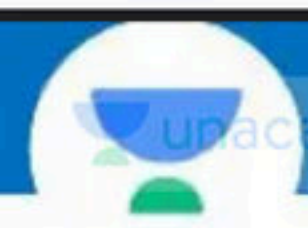
(1) I, II, III and VI

(2) III, IV, V and VI

(3) I, II, V and VI

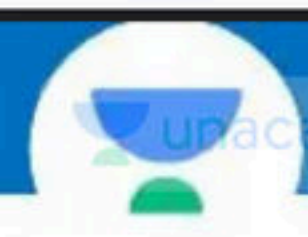
✓ (4) I, II, III, IV and V





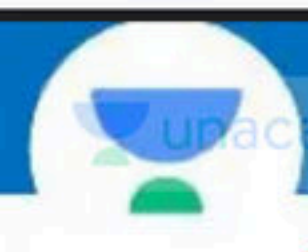
20. Which of the following option does not include the antagoistic hormones:

- (1) Insulin and glucagon
- (2) PTH and calcitonin
- ✓ (3) Epinephrine and norepinephrine → same
- (4) All are antagonistic hormones



**21. 'Mammalian thymus' is mainly concerned with :**

- (1) Regulation of body temperature
- (2) Regulation of body growth
- ✓(3) Immunological functions ✓
- (4) Secretion of thyrotropin



### 22. Due to insulin secretion :

- (1) Increase in the level of ketone bodies in the blood
- (2) Increase in the level of glucose in the blood
- (3) Increase in level of glucose and ketone body in blood
- ✓ (4) Decrease in the level of glucose in the blood





**23.** Match the hormone with the roles given under.

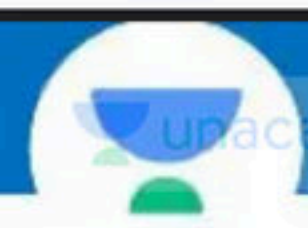
Select the choice in which the alphabets of the two columns are correctly matched :

**Hormones****Roles**

- |                 |                                                |
|-----------------|------------------------------------------------|
| a. FSH          | i. Preparation of endometrium for implantation |
| b. LH           | ii. Female secondary sexual characters         |
| c. Progesterone | iii. Contraction of uterine muscles = oxytocin |
| d. Estrogen     | iv. Development of corpus luteum               |
|                 | v. Maturation of Graafian follicle             |

(1) a-ii, b-iv, c-i, d-iii      (2) a-v, b-i, c-iv, d-ii

(3) a-iii, b-v, c-iv, d-ii      (4) a-v, b-iv, c-i, d-ii



**24.** Match the following columns :

### Column I

- a. Zona reticularis
- b. Zona fasciculata
- c. Zona glomerulosa

### Column II

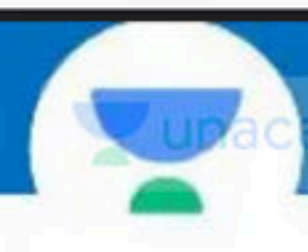
- i. Outer layer (adrenal cortex)
- ii. Inner layer (adrenal cortex)
- iii. Middle layer (adrenal cortex)

(1) a-iii, b-ii, c-i

(2) a-i, b-ii, c-iii

☒ (3) a-ii, b-iii, c-i

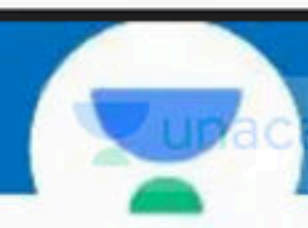
(4) a-ii, b-i, c-iii



25. Which of the option shows hormone involve in carbohydrate metabolism:

- (1) Insulin, glucagon, progesterone, estrogen
- (2) Progesterone, estrogen
- (3) Glucocorticoids, Oxytocin, epinephrine
- ✓ (4) Insulin, gulcagon, Glucocorticoids, epinephrine





**26. Given below are two statements :**

✓ **Statement I :**

Prolonged hyperglycemia lead to a complex disorder called diabetes mellitus.

✓ **Statement II :**

Diabetes mellitus patients are successfully treated with insulin therapy.

**Choose the correct answer from the options given below**

- (1) Both Statement I and Statement II are incorrect
- (2) Statement I is correct but Statement II is incorrect
- (3) Statement I is incorrect but Statement II is correct
- ✓ (4) Both, Statement I and Statement II are correct



27. Given below are two statements :

**Statement I :**

Glucoc.

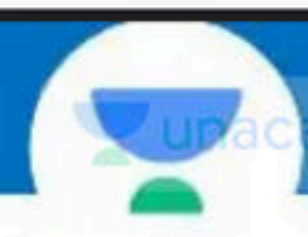
In our body cortisol is main mineralocorticoid. ✗

**Statement II :**

Cortisol involved in maintaining the cardio-vascular system as well as the kidney function.

**Choose the correct answer from the options given below**

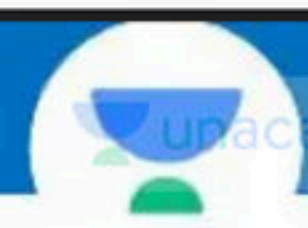
- (1) Both Statement I and Statement II are incorrect
- (2) Statement I is correct but Statement II is incorrect
- ✓(3) Statement I is incorrect but Statement II is correct
- (4) Both, Statement I and Statement II are correct



**28. Graves' disease is caused due to :-**

- (1) hyposecretion of thyroid gland
- ✓ (2) hypersecretion of thyroid gland
- (3) hyposecretion of adrenal gland
- (4) hypersecretion of adrenal gland.





29. Match the hormone with its source

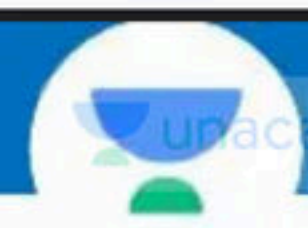
- |                 |                         |
|-----------------|-------------------------|
| a. somatostatin | i. Pineal gland         |
| b. Melatonin    | ii. Corpus luteum       |
| c. Aldosterone  | iii. Placenta           |
| d. Progesterone | iv. Adrenal cortex      |
| e. HCG          | v. Islets of Langerhans |
|                 | vi. Adenohypophysis     |
- 

(1) a-v, b-i, c-vi, d-iii, e-ii

(2) a-i, b-ii, c-iv, d-iii, e-v

(3) a-ii, b-vi, c-iv, d-v, e-iii

✓ (4) a-v, b-i, c-iv, d-ii, e-iii



30. Given below are two statements :

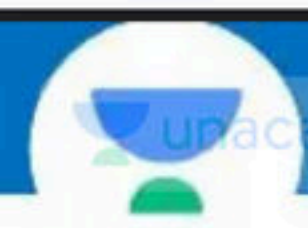
**Statement I :** Thymosin play a major role in differentiation of Tlymphocytes which provide cell metiated immunity. ✓

**Statement-II :**

Thymosin promote production of antibodies to provide humoral Immunity ✓

**Choose the correct answer from the option given below:**

- (1) Both Statement I and Statement II are incorrect
- (2) Statement I is correct but Statement II is incorrect
- (3) Statement I is incorrect but Statement II is correct
- ✓ (4) Both, Statement I and Statement II are correct



31. Which of the following hormone stimulates the synthesis and secretion of steroid hormones called glucocorticoids from the adrenal cortex

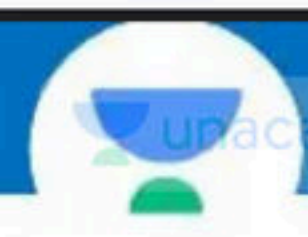
(1) GH ↓

✓(2) ACTH

(3) Prolactin

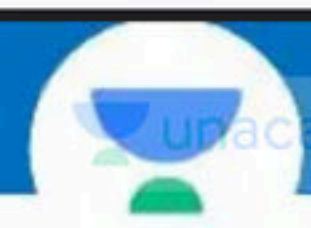
(4) TSH





**32. In males, LH stimulates the synthesis and secretion of :**

- ✓(1) Androgens from testis ↪
- (2) Progesteron from ovary
- (3) Estrogen from ovary
- (4) Ovulation from ovary



33. Given below are two statements :

**Statement I :**

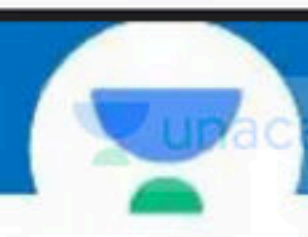
The juxtaglomerular cells of kidney produce a peptide hormone called Renin which stimulates erythropoiesis ✗ erythropoietin

**Statement-II :**

GIP Inhibits gastric secretion and Motility. ✓

**Choose the correct answer from the option given below:**

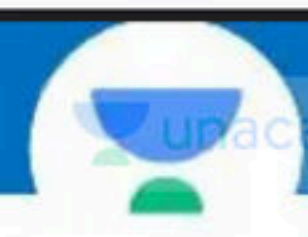
- (1) Both Statement I and Statement II are incorrect
- (2) Statement I is correct but Statement II is incorrect
- ✓ (3) Statement I is incorrect but Statement II is correct
- (4) Both Statement I and Statement II are correct



**34. Which one is essential for the hormone synthesis in the thyroid.**

- (1) Sodium
- (2) Iron
- (3) Chlorine
- ☒ (4) Iodine





**35** Which hormone stimulates the RBC production

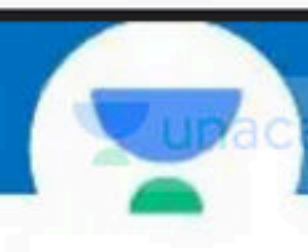
(1) Vasopressin

(2) ADH

(3) ACTH

☒ (4) Cortisol

↓  
cortisol



### PART-1 (SECTION-B)

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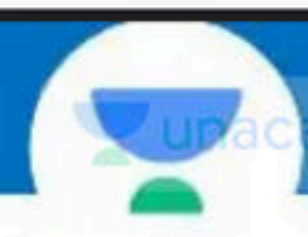
**36. Which one is acts as both exocrine and endocrine gland**

(1) Parathyroid

(2) Testis

(3) Adrenal

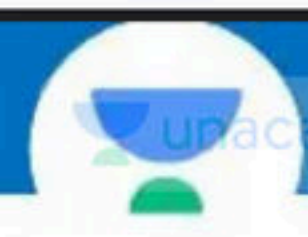
✓ (4) Pancreas



37. Which statement is incorrect for insulin hormone :-

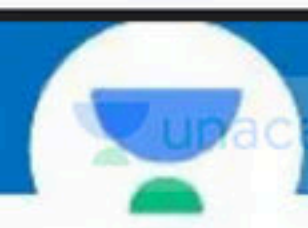
- (1) Acts mainly on hepatocytes ✓
- ✓ (2) It is a <sup>Peptide</sup> ~~steroid~~ hormone
- (3) Enhances cellular glucose uptake and utilisation ✓
- (4) None of these





**38. Diabetes mellitus patients are successfully treated with :**

- ✓ (1) Insulin ↪
- (2) Vasopressin
- (3) Glucagon
- (4) ADH



**39. Match the following columns and select the correct option**

**Column - I**

**Column - II**

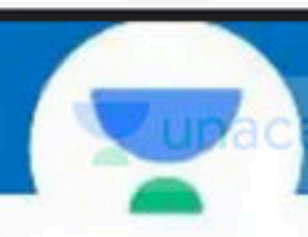
- |                     |                          |
|---------------------|--------------------------|
| (A) Pituitary gland | (i) Grave's disease      |
| (B) Thyroid gland   | (ii) Diabetes mellitus   |
| (C) Adrenal gland   | (iii) Diabetes insipidus |
| (D) Pancreas        | (iv) Addison's disease   |
- 

(1) A-iv, B-iii, C-i, D-ii

(2) A-iii, B-ii, C-i, D-iv

✓ (3) A-iii, B-i, C-iv, D-ii

(4) A-ii, B-i, C-iv, D-iii

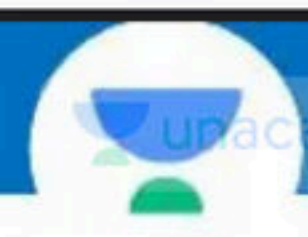


40 Which is a wrong statement regarding adrenal cortex:

- (a) Epinephrine stimulates sympathetic nervous system  $\rightarrow$  medulla
- (b) Cortisol is a hypoglycemic hormone  $\rightarrow$  X
- (c) Cortisol is a glucocorticoid
- (d) Cortisol suppresses the immune response

- ✓(1) a and b
- (2) b and c
- (3) c and d
- (4) a and d





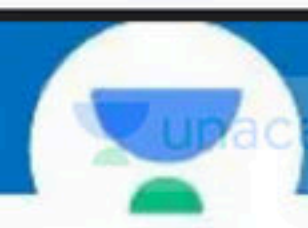
**41.** The hormone which is responsible for the ovulation:

(1) FSH

☒ (2) LH

(3) PRL

(4) MSH



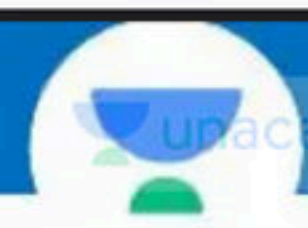
42. Which one of the following pairs of hormones are the examples of those that can easily pass through the cell membrane of the target cell and bind to a receptor inside it :-

(1) Somatostatin, oxytocin

✓ (2) Cortisol, testosterone = steroid hormone

(3) Insulin, glucagon

(4) Thyroxin, Insulin



43 Which is a correct matching set :

Column I

a- Glucagon

b- Cortisol

c- Fear or fight

d- Insulin

Column II

(i) Epinephrine

(ii) Hypoglycemia

(iii) Glycogenolyses hormone

(iv) Hyperglycemia

✓ (1) a-iii, b-iv, c-i, d-ii

(2) a-iii, b-iv, c-ii, d-i

(3) a-iv, b-iii, c-i, d-ii

(4) a-iii, b-i, c-iv, d-ii



44. Which is a correct matching set :

Column I

Column II

a. Hypothalamus

p. Insulin

b. Anterior pituitary

q. Estrogen

c. Testis

r. FSH and LH

d. Ovary

s. Androgens

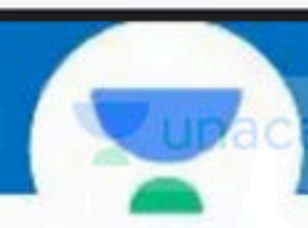
t. Gonadotropin releasing hormone

~~(1)~~ a-t, b-r, c-s, d-q

(2) a-t, b-r, c-q, d-s

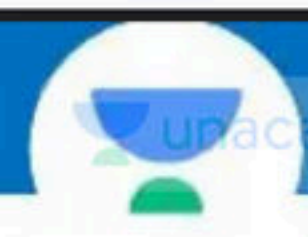
(3) a-p, b-q, c-s, d-r

(4) a-r, b-t, c-s, d-q



45. The pituitary gland is divided anatomically into :

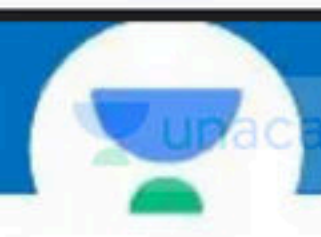
- (1) An adenohypophysis and pars intermedia
- (2) An adenohypophysis and a epiphysis
- (3) An epiphysis and a neurohypophysis
- (4) An adenohypophysis and a neurohypophysis.



**46.** Pars distalis and pars intermedia is part of

- ☒ (1) An adenohypophysis
- (2) An adenohypophysis and a neurohypophysis
- (3) An epiphysis
- (4) Neurohypophysis





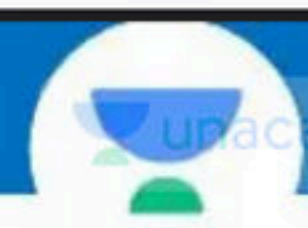
47. Pars intermedia secretes a hormone called

☒ (1) MSH

(2) ACTH

(3) FSH

(4) GH



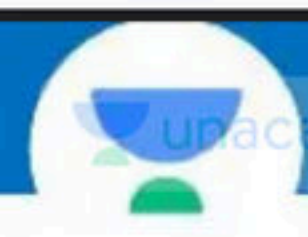
48. <sup>TSH</sup>..... stimulates the synthesis and secretion of thyroid hormones from the thyroid gland.

(1) GH

(2) ACTH

(3) Prolactin

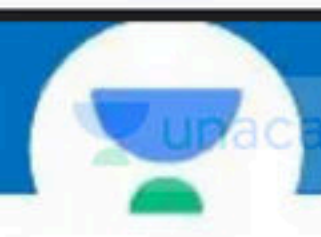
✓ (4) TSH



**49. Most abundant and phagocytic leucocyte is**

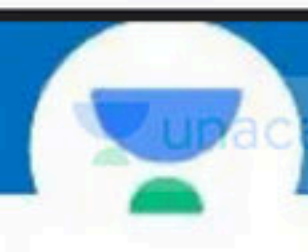
- (1) Neutrophils and basophils
- (2) Eosinophils and basophils
- (3) Basophils and neutrophils respectively
- ✓ (4) Neutrophils and neutrophils respectively





50. *Chordae Tendinae* are found in :

- ✓ (1) Ventricles of heart
- (2) Atria of heart
- (3) Both ventricle and atria
- (4) Only right ventricle of heart

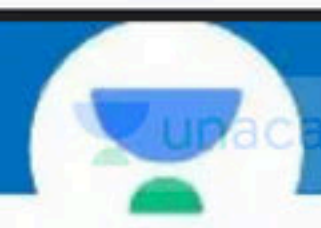


### PART-2 (SECTION-A)

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**51. QRS complex represent the :**

- (1) Repolarization of ventricles
- ✓(2) Depolarization of ventricles ✓
- (3) Repolarization of atria
- (4) Depolarization of atria



52. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R)

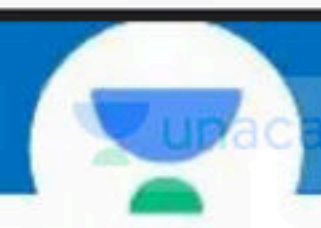
✓ Assertion (A) : Thrombins in turn are formed from inactive substance present in plasma called prothrombin.

✓ Reason (R) : An enzyme complex thrombokinese required for changing prothrombin to thrombin.

Choose the correct answer from the options given below :

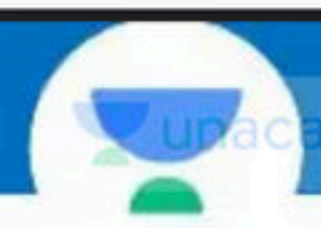
- (1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- (2) (A) is correct but (R) is not correct
- (3) (A) is not correct but (R) is correct
- ✓ (4) Both (A) and (R) are correct and (R) is the correct explanation of (A)





53. Name the blood cells, whose reduction in number can cause clotting disorder, leading to excessive loss of blood from the body :

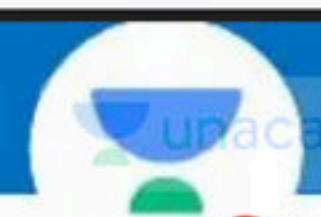
- (1) Erythrocytes
- (2) Leucocytes
- (3) Neutrophils
- ✓ (4) Thrombocytes



54. What would be the heart rate of parson if the cardiac output is 5 L. blood volume in the ventricles at the end of diastole is 100 ml. and at the end of ventricular systole is 50 mL :

- (1) 50 beat per minute
- (2) 75 beat per minute
- ✓ (3) 100 beat per minute
- (4) 125 beat per minute

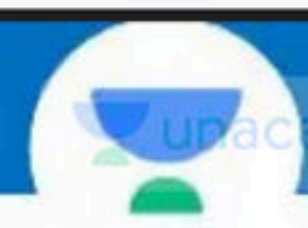
$$\frac{5000}{50} = 100$$



**55. Plasma contains which of the following minerals**

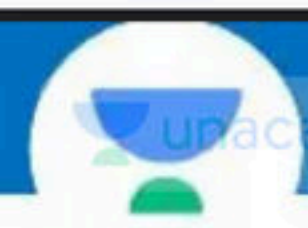
- (1) Sodium, calcium ions ✓
- (2) Magnesium, bicarbonate ions ✓
- (3) Chloride ions ✓
- ✓ (4) All of these





56. Heart → deoxygenated blood → gills → oxygenated blood → body parts → deoxygenated blood → heart. This is

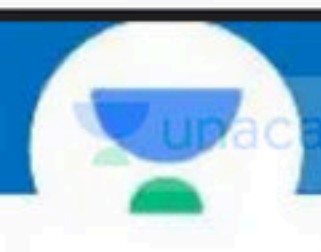
- ✓ (1) Single circulation
- (2) Incomplete double circulation
- (3) Double circulation
- (4) None of these



57. The flow of blood in human heart is from

- (1) Ventricle Atria Pulmonary artery Atria
- (2) Atria Pulmonary artery Ventricle Atria
- ✓ (3) Atria Ventricle Pulmonary artery → Lungs
- (4) None of these

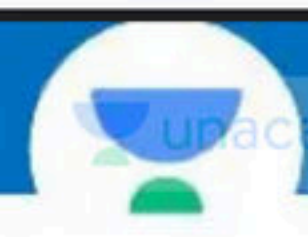
A → V → P.A



58. Which of the following represents systemic circulation :-
- (1) Right Atrium Dorsal aorta Tissue Vena cava Left atrium
  - (2) Left Atrium Vena cava Tissue Dorsal aorta Right atrium
  - ✓ (3) LA Dorsal aorta Tissue Vena cava Right atrium
  - (4) LA Dorsal aorta Lungs Vena cava Right atrium

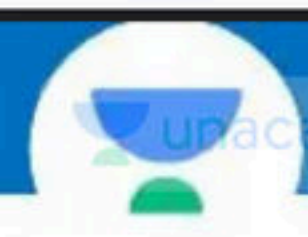
LA → D.A → Tissue → Ven. →  
R.A





### 59. Function of systemic circulation include

- (1) Provides nutrients and Oxygen supply to tissue ✓
- (2) Takes CO<sub>2</sub> and harmful substances from tissue away for elimination ✓
- (3) Takes nutrients and oxygen from tissue
- ✓ (4) Both 1 and 2



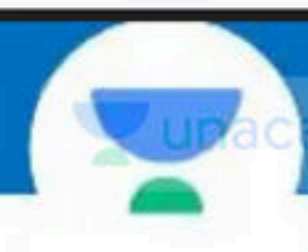
**60** Mark the odd one out :

(1) Lymphocytes

(2) Monocytes

☒ (3) Erythrocytes

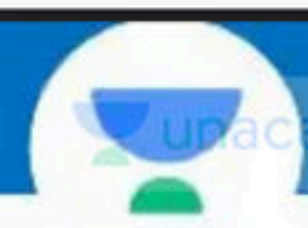
(4) Neutrophils



**61** Bicuspid valve found between :

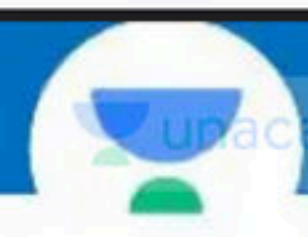
- (1) The right atrium and the right ventricle
- ✓ (2) The left atrium and the left ventricle
- (3) The right atrium and the left ventricle
- (4) The left atrium and the right ventricle





62. Which is a correct statement :-

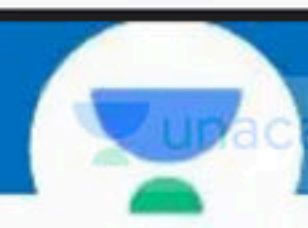
- (a) Lymph and blood are body fluid ✓
  - (b) Purkinje fibres are present in heart of Hydra ✗
  - (c) Cockroach has <sup>open</sup>~~closed~~ vascular system ✗
  - (d) Coronary artery supplies blood to heart muscles ✓
- (1) a and b
  - (2) b and c
  - ✓ (3) a and d
  - (4) b and d



**63.** Which is a wrong statement

- (a) E.C.G. has clinical importance ✓
- (b) Adrenal medullary hormone has no effect on cardiac output ✗
- (c) Spleen is the graveyard of RBC in body ✓
- (d) Atherosclerosis is thining of arterial wall ✗

- (1) Only a and b
- (2) Only b and c
- ✓✓ (3) Only b and d
- (4) Only d



**64** Given below are two statements :

**Statement I : Persons with blood group AB can donate blood to person with same blood group under ABO system**

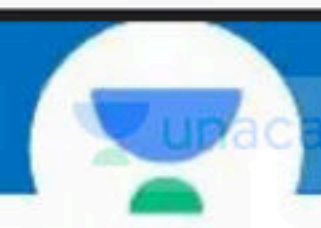
**Statement-II :**

**Calcium ions play a very important role in blood clotting**

**Choose the correct answer from the option given below:**

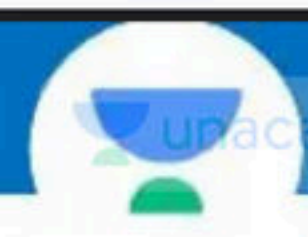
- (1) Both Statement I and Statement II are incorrect
- (2) Statement I is correct but Statement II is incorrect
- (3) Statement I is incorrect but Statement II is correct
- ☒ (4) Both Statement I and Statement II are correct





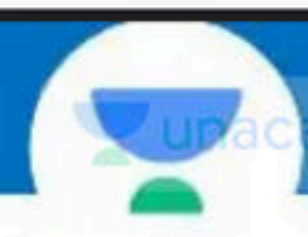
65. Which of the following type cells lack nucleus in most of mammals :

- ✓ (1) Erythrocytes
- (2) Eosinophils
- (3) Neutrophils
- (4) Monocyte



**66.** Which one of the following is correct :

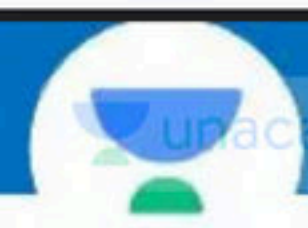
- ✓ (1) Blood = Plasma + RBC+WBC+Platelets
- (2) Plasma = Blood + Lymphocytes
- (3) Serum = Blood + Fibrinogen
- (4) Lymph = Plasma + RBC + WBC



**67. Serum differs from blood in :**

- ✓ (1) Lacking clotting factors ↺
- (2) Lacking antibodies
- (3) Lacking globulins
- (4) Lacking albumins





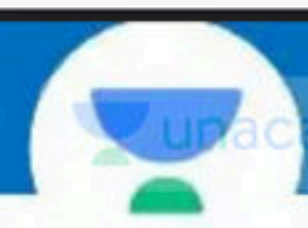
68. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R)

**Assertion (A) :** Renin converts angiotensinogen to angiotensin I and further to angiotensin II which increase G.F.R. ✓

**Reason (R) :** Anigotensin II, Increase the glomerular blood pressure and also activate the adrenal <sup>105FA</sup>medulla to release aldosterone and there by G.F.R. ✗

**choose the correct answer from the options given below :**

- (1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- ✓(2) (A) is correct but (R) is not correct
- (3) (A) is not correct but (R) is correct
- (4) Both (A) and (R) are correct and (R) is the correct explanation of (A)



69. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R)

✓ **Assertion (A)** : Tubular secretion helps in the maintenance of ionic and acid-base balance of body fluids.

✓ **Reason (R)** : During urine formation the tubular cells secrete substances like  $H^+$ ,  $K^+$  and ammonia in to filtrate.

**choose the correct answer from the options given below :**

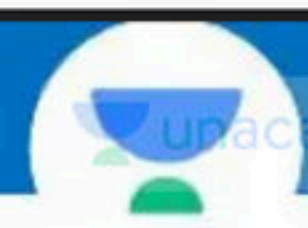
(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(2) (A) is correct but (R) is not correct

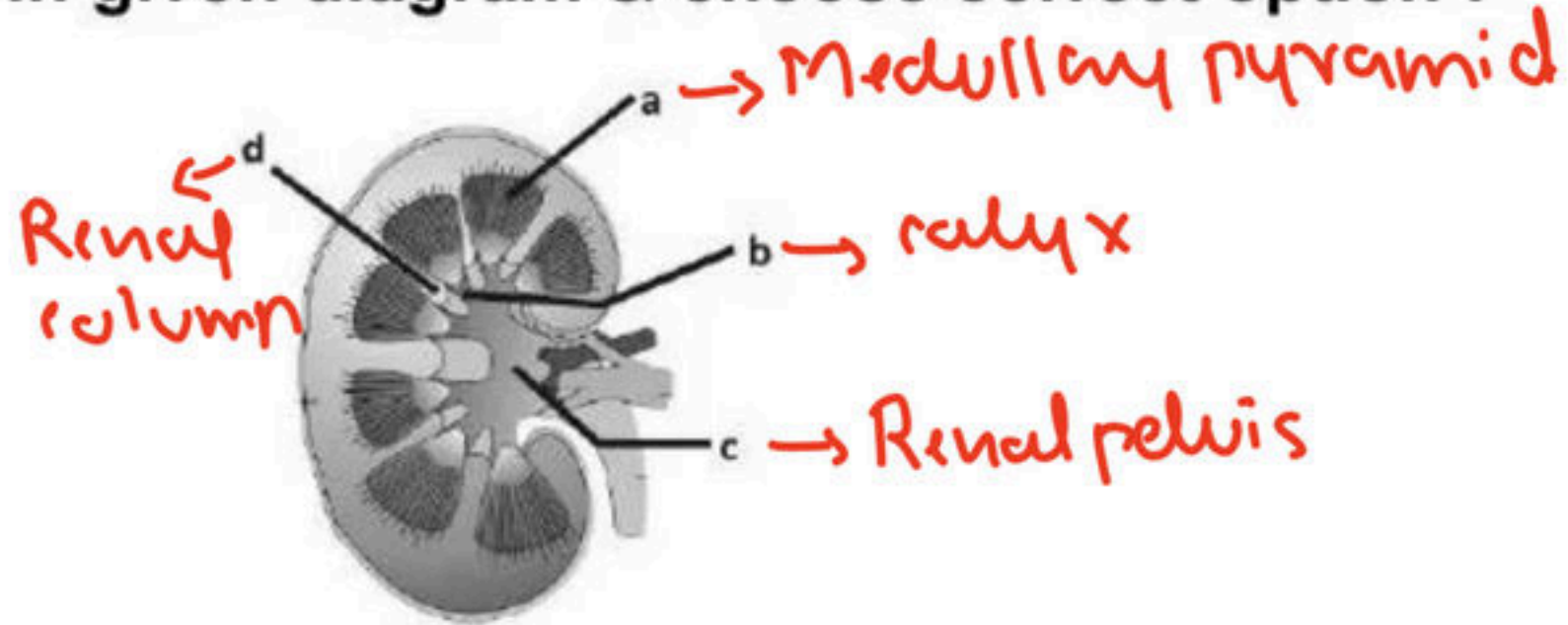
(3) (A) is not correct but (R) is correct

✓✓ (4) Both (A) and (R) are correct and (R) is the correct explanation of (A)



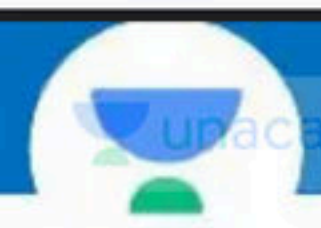


70. Identify a, b, c and d in given diagram & choose correct option :



- (1) a-Renal column, b-Medullary pyramid
- (2) c-Calyx, d-Medullary pyramid
- (3) b-Renal column, c-Renal pelvis
- ✓ (4) a-Medullary pyramid, d- Renal column

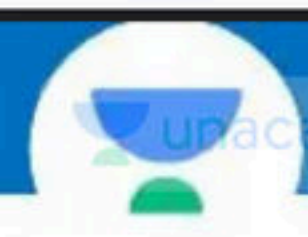




71. The correct sequence of meninges of spinal-cord from inside to outside is:

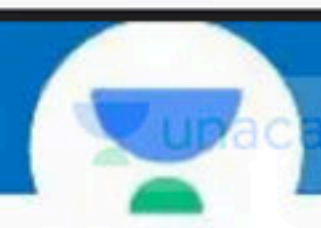
- (1) Dura mater Arachnoid Pia mater
- (2) Arachnoid Dura mater Pia mater
- ✓(3) Pia mater Arachnoid Dura mater
- (4) Dura mater Pia mater Arachnoid

1 pad



72. Which part of human brain is concerned with the regulation of body temperature :

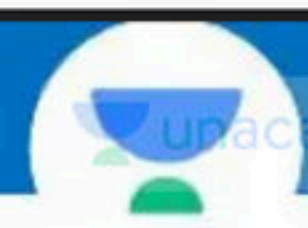
- (1) Medulla oblongata
- ✓ (2) Hypothalamus
- (3) Cerebrum
- (4) Cerebellum



73. Injury localized to the <sup>"</sup>hypothalamus<sup>"</sup> would most likely disrupt :

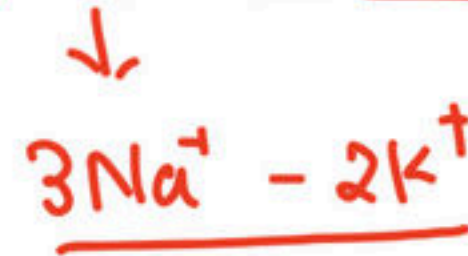
- (1) Short - term memory
- (2) Co-ordination during locomotion
- (3) Executive functions, such as decision making
- ✓ (4) Regulation of body temperature.

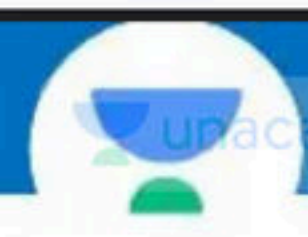




**74** The ionic gradients across the resting membrane are maintained by the active transport of ions by the sodium-potassium pump which transports

- (1) 3 Na<sup>+</sup> outwards for 1K<sup>+</sup> into the cell.
- (2) 3 Na<sup>+</sup> outwards for 3K<sup>+</sup> into the cell.
- ✓ (3) 3 Na<sup>+</sup> outwards for 2K<sup>+</sup> into the cell.
- (4) 2 Na<sup>+</sup> outwards for 2K<sup>+</sup> into the cell.





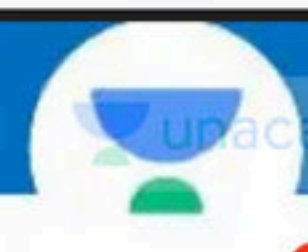
75. Bipolar neuron found in

(1) Cerebral cortex → *Multipolar*

✓(2) Retina of eye

(3) Hydra

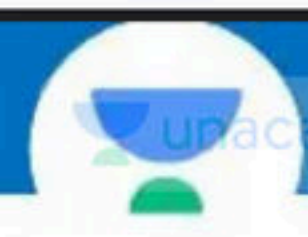
(4) All of these



**76. Impulse transmission across an electrical synapse is :**

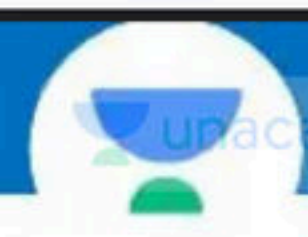
- (1) Always same as that of a chemical synapse.
- (2) Sometime faster than that of a chemical synapse.
- (3) Always slower than that of a chemical synapse.
- ✓ (4) Always faster than that of a chemical synapse.





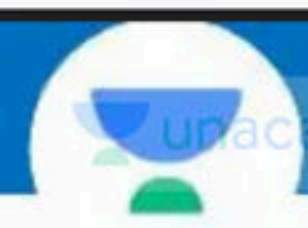
77. The medulla contains centres which control :-

- (1) Respiration ✓
- (2) Cardiovascular reflexes ✓
- (3) Gastric secretions ✓
- (4) ✓ All of the above



**78.** You have two types of brain in petridish. How would you distinguish brain of mammal. By :-

- ✓ (1) Corpus callosum
- (2) Optic lobes
- (3) Cerebellum
- (4) Cerebrum



79. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R)

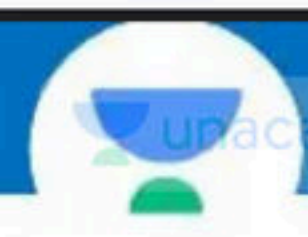
**Assertion:** Unmyelinated nerve fibre is enclosed by Schwann cell

**Reason:** Schwann cell are <sup>-1st</sup> absent in myelinated nerve fibre. ✗

In the light of the above statements, choose the correct answer from the options given below :

- (1) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- ✓ (2) (A) is true but (R) is false
- (3) (A) is false but (R) is true
- (4) Both (A) and (R) are true and (R) is the correct explanation of (A)



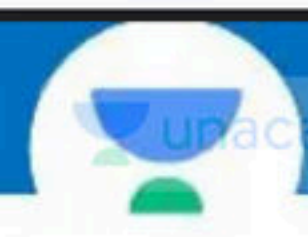


80. The conduction of nerve impulse is a :-

- (1) Biochemical phenomenon
- ✓ (2) Electrochemical phenomenon
- (3) Physical phenomenon
- (4) Bio-physiological phenomenon

81. The cerebrum is <sup>long.</sup> \_\_\_A\_\_\_ divided into two halves that are connected by <sup>corpus call.</sup> \_\_\_B\_\_\_.

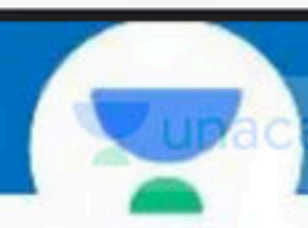
- (1) A—Longitudinally, B—Corpus callum
- ✓ (2) A—Longitudinally, B—Corpus callosum
- (3) A—Transverselly, B—Corpus callum
- (4) A—Transverselly, B—Corpus callosum



82. Nerve impulse is generated when the nerve cell undergoes or passes :

- (1) Excited stage ✓
- (2) Depolarization ✓
- (3) Action potential ✓
- ✓ (4) All of these

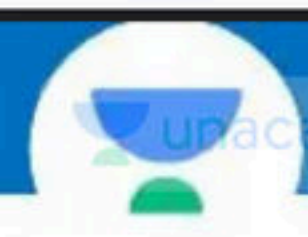




83. Which of the following is not correctly matched :

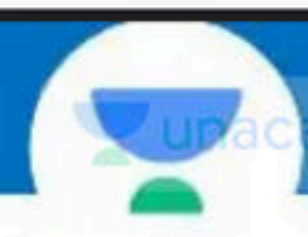
- (1) Cerebellum - Balance ✓
- (2) Limbic system - Regulation of sexual behaviour ✓
- (3) Hypothalamus - Urge for eating ✓
- ✓ (4) Medulla oblongata - Temperature regulation

↓ x  
hypoth



**84. The human hind brain comprises three parts, one of which is**

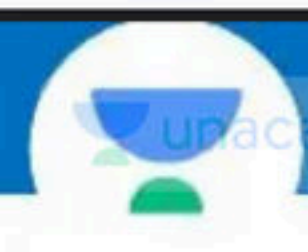
- ☒ (1) Cerebellum
- (2) Hypothalamus
- (3) Spinal
- (4) Corpus callosum



**85. Nissl's granules are present in :**

- (1) Cyton ✓
- (2) Axon
- (3) Dendron ✓
- ✓ (4) Both 1 and 3

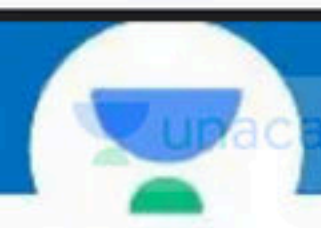




### PART-2 (SECTION-B)

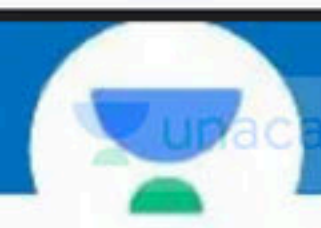
**86.** Which cranial meninges is in direct contact with brain :

- (1) Duramater
- (2) Arachnoid
- ✓ (3) Piamater
- (4) Both 1 and 3



87. Which of the following neural system relays the impulse from CNS to skeletal muscles :

- ↓
- ✓ (1) Somatic neural system ✓
  - (2) Sympathetic neural system
  - (3) Parasympathetic neural system
  - (4) Autonomic neural system



88. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R)

**Assertion (A) :** The PNS comprises all the nerves of the body associated with the CNS. ✓

**Reason (R) :** The PNS is further classified in to SNS and ~~PNS~~ ×

ANS

**Choose the correct answer from the options given below :**

- (1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- ✓(2) (A) is correct but (R) is not correct
- (3) (A) is not correct but (R) is correct
- (4) Both (A) and (R) are correct and (R) is the correct explanation of (A)





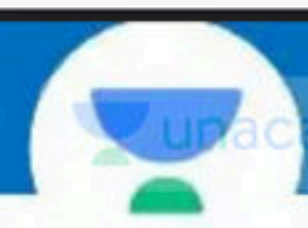
**89. Given below are two statements :**

**Statement I :** Electrical synapses are rare in our system.

**Statement II :** Chemicals called neurotransmitters are involved in the transmission of impulse at chemical synapses.

**Choose the correct answer from the options given below**

- (1) Both Statement I and Statement II are incorrect
- (2) Statement I is correct but Statement II is incorrect
- (3) Statement I is incorrect but Statement II is correct
- ☒ (4) Both, Statement I and Statement II are correct



**90. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R)**

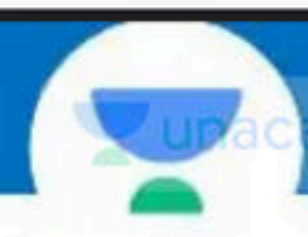
**Assertion (A) :** The immune responses of old persons become weak.

**Reason (R) :** Thymus is degenerated in old, individual resulting in a decreased production of thymosins.

**Choose the correct answer from the options given below :**

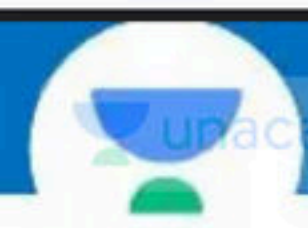
- (1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- (2) (A) is correct but (R) is not correct
- (3) (A) is not correct but (R) is correct
- ✓(4) Both (A) and (R) are correct and (R) is the correct explanation of (A)

**91. Glomerulus alongwith Bowman's capsule is called:**



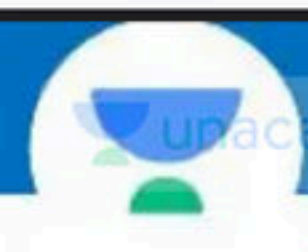
- (1) Blood corpuscle
- (2) Malpighian ~~tubule~~ *Body*
- ✓ (3) Renal corpuscle
- (4) Both 2 & 3





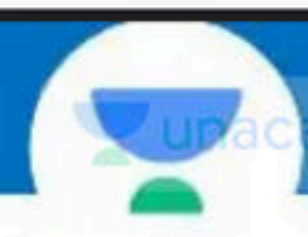
92. A minute vessel runs parallel to the Henle's loop of juxta medullary nephrons called :

- (1) Efferent arteriole
- (2) Renal vein
- ~~✓~~ (3) Vasa recta
- (4) Dorsal aorta



**93** Which statement is incorrect :

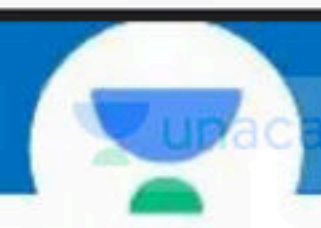
- (1) In ascending limb of loop of henle reabsorption is minimum  
*concl.*
- ✓ (2) ~~Dilution~~ of urine done by juxta medullary nephrons
- (3) The ascending limb of loop of Henle is impermeable to water
- (4) PCT is lined by simple cuboidal brush border epithelium which increases surface area for reabsorption



94. NaCl from ascending Henle's loop is exchanged with :-

- (1) Ascending limb of vasa recta
- ✓ (2) Descending limb of vasa recta
- (3) Afferent arteriole
- (4) Descending limb of Henle's loop

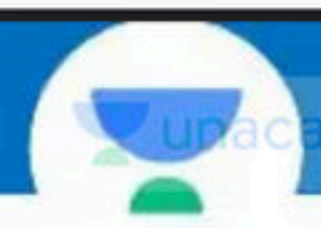




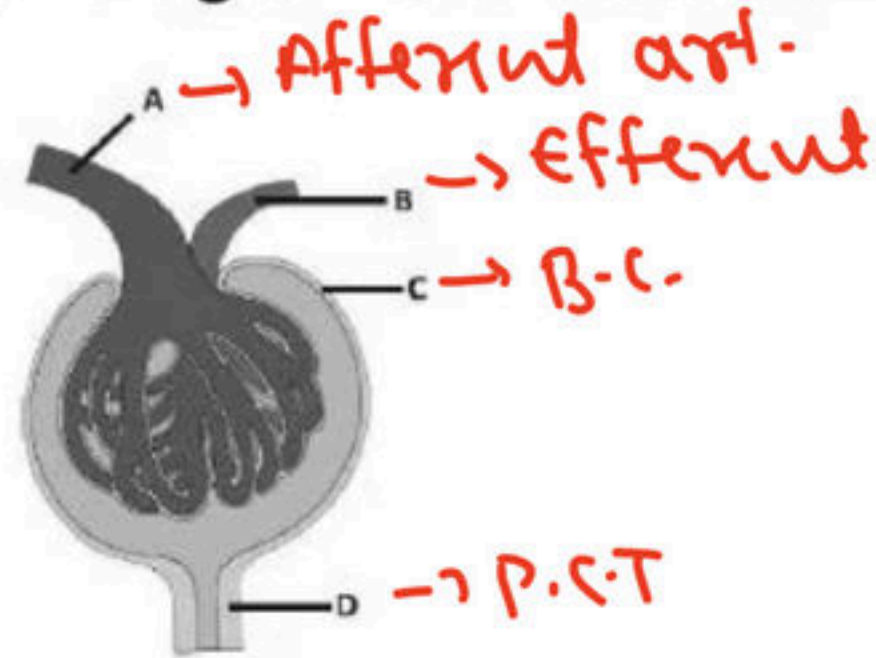
**95. The thickness of each kidney of an adult human is about:**

- (1) 14-15 cm
- (2) 10-12 cm
- (3) 21-25 cm
- (4) 2-3 cm

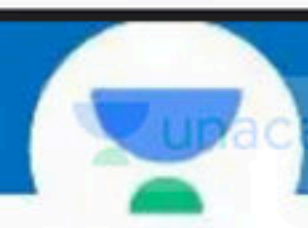




96 Identify A, B, C and D in given diagram & choose correct option :



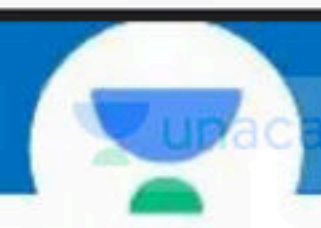
- (1) A–Afferent arteriole, B–Bowman's capsule
- (2) C–PCT, D–Efferent arteriole
- ✓ (3) B–Efferent arteriole, C–Bowman's capsule
- (4) A–PCT, D–Afferent arteriole



**97** ADH is :

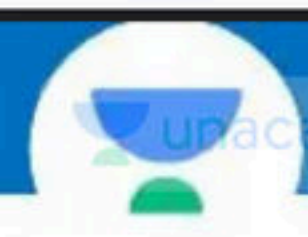
- (1) Enzyme that hydrolyses peptides
- ✓ (2) Hormone released by pituitary that promotes reabsorption of water from glomerular filtrate
- (3) Hormone that promotes glycogenolysis
- (4) Energy rich compound connected with muscle contraction.





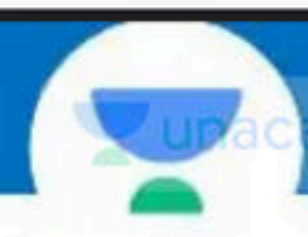
**98. Which excretory structures are found in rotifers:**

- (1) Metanephridia
- (2) Kidney
- ✓ (3) Protonephridia
- (4) Antennal gland



99. Inner to the hilum is a broad funnel shaped space called:

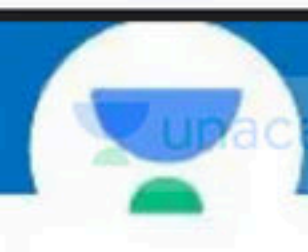
- (1) Ureter
- ✓(2) Renal pelvis
- (3) Urinary bladder
- (4) Cortex



11 11  
100. DCT is also capable of selective secretion :

- (1) Hydrogen ion ✓
- (2) Potassium ion ✓
- (3)  $\text{HCO}_3^-$
- ✓ (4) Both 1 & 2

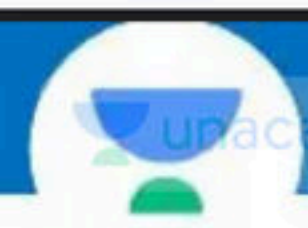




## PART-3 (SECTION-A)

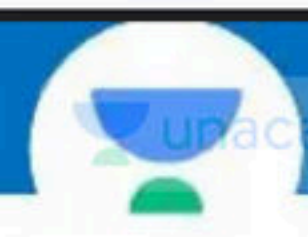
**101. Which statement is incorrect :**

- (1) ADH facilitates water reabsorption from latter part of tubule
- (2) ANF mechanism act as a check on the renin angiotensin mechanism
- ✓ (3) Atrial Natriuretic Factor activates the adrenal cortex to release aldosterone ✗
- (4) Presence of glucose in urine are indicative of diabetes mellitus.



102. In glomerulus blood filters through how many  
layers

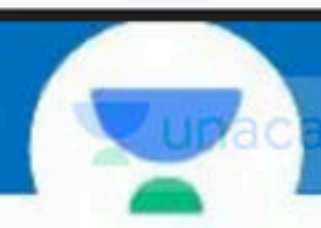
- (1) One
- ✓ (2) Three
- (3) Two
- (4) Four



**103** A drop in GFR is regulated by secretion of which of the following hormone by Juxta glomerular cells : Renin

- ☒ (1) Renin
- (2) Prolactin
- (3) Trypsin
- (4) Pepsin



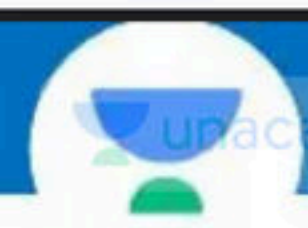


104. Given below are two statements :

- ☒ **Statement I :** JGA is special sensitive region formed by cellular modification distal convoluted tubule and the efferent arteriole at location of their contact.
- ☒ **Statement-II :** ANF is a vasodilator and can decrease blood pressure.

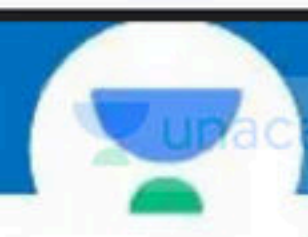
**Choose the correct answer from the option given below:**

- (1) Both Statement I and Statement II are incorrect
- (2) Statement I is correct but Statement II is incorrect
- ☒ (3) Statement I is incorrect but Statement II is correct
- (4) Both Statement I and Statement II are correct



105. The osmolarity in cortex and inner medullary interstitial space is

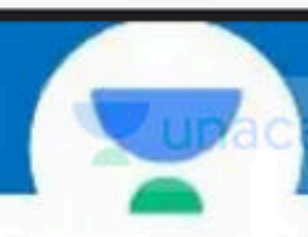
- (1) 300 and 500 mosmolL-
- ✓ (2) 300 and 1200 mosmolL-
- (3) 1200 and 300 mosmolL-
- (4) 1200 and 800 mosmolL-



**106.** Atrial natriuretic factor is released from :-

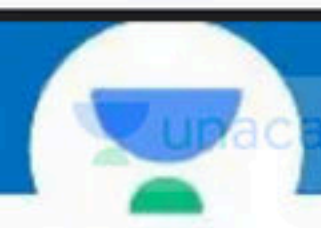
- ✓ (1) Heart ↩
- (2) Kidney
- (3) Lungs
- (4) Brain





**107** The vessel leading blood (containing nitrogenous waste)  
into the Bowman's capsule is known as

- ✓ (1) Afferent arteriole ↩
- (2) Efferent arteriole
- (3) Renal artery
- (4) Renal vein



**108** Arrange the parts of the nephron in a correct sequence.

(I) Glomerulus

(II) Bowman's capsule

(III) Henle's loop

(IV) Proximal convoluted tubule

(V) Distal convoluted tubule

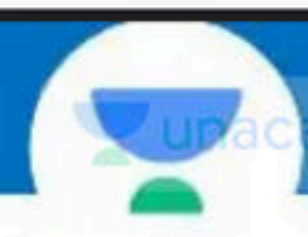
(1) I II III IV V

✓(2) I II IV III V

(3) I II V III IV

(4) IV III II I V

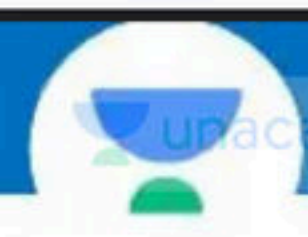
I, II, IV, III, V



**109. Which statement is correct :**

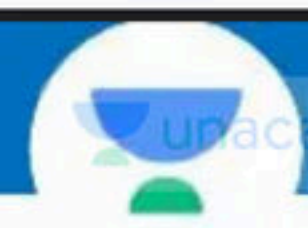
- (1) GFR in a healthy individual is approximately 125 ml/minute.
- (2) The urine released per day is approx 1.5 litre
- (3) In PCT, 70-80 per cent of electrolytes and water are reabsorbed
- ✓ (4) All of these





**110. Tell the wrong pair**

- (1) Bony fishes ammonotelic
- ✓ (2) Aquatic insects ~~uricotelic~~ *ammonotelic*
- (3) Mammals ureotelic
- (4) Reptiles uricotelic



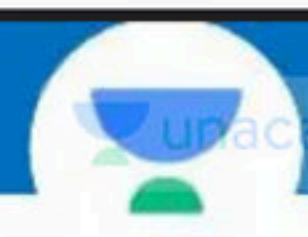
**111.** Given below are two statements :

**Statement I :** Presence of glucose and ketone bodies in urine are indicative of diabetes mellitus.

**Statement II :** Prolonged hyper glycemia lead to a complex disorder called diabetes mellitus.

**Choose the correct answer from the options given below**

- (1) Both Statement I and Statement II are incorrect
- (2) Statement I is correct but Statement II is incorrect
- (3) Statement I is incorrect but Statement II is correct
- ☒ (4) Both, Statement I and Statement II are correct



**112. Match the item given in column I with those in column II select the correct option given below :**

- |                              |                                                         |
|------------------------------|---------------------------------------------------------|
| <b>a) Glycosuria</b>         | <b>i) Accumulation of uric acids in joints</b>          |
| <b>b) Gout</b>               | <b>ii) Mass of crystallised salts within the kidney</b> |
| <b>c) Renal calculi</b>      | <b>iii) Inflammation of glomeruli</b>                   |
| <b>d) Glomerulonephritis</b> | <b>iv) Presence of glucose in urine</b>                 |
- 

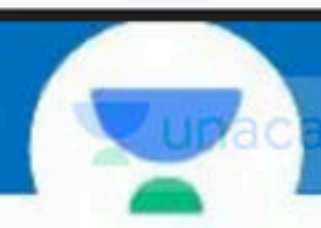
(1) a-iii, b-ii, c-iv, d- i

(2) a-iv, b-i, c-iii, d- ii

(3) a-i, b-ii, c-iii, d- iv

☒ (4) a-iv, b-i, c-ii, d- iii





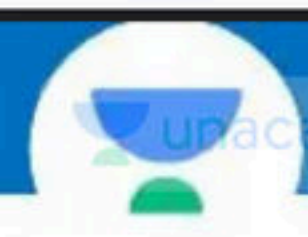
**113.** Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R)

Assertion (A) : JGA play a complex regulatory role on kidney. ✓

Reason (R) : A fall in GFR can activate the JG cells to release <sup>renin</sup> ~~angiotensin~~. ✗

Choose the correct answer from the options given below :

- (1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- ✓ (2) (A) is correct but (R) is not correct
- (3) (A) is not correct but (R) is correct
- (4) Both (A) and (R) are correct and (R) is the correct explanation of (A)



114. Descending loop of henle is:

- a. Permeable to water ✓
- b. Impermeable to electrolytes ✓
- c. Impermeable to water
- d. Permeable to electrolytes

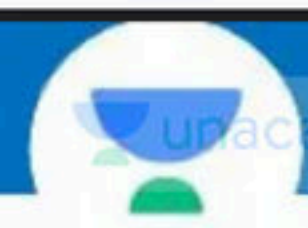
✓ (1) a, b are true

(2) c,d are true

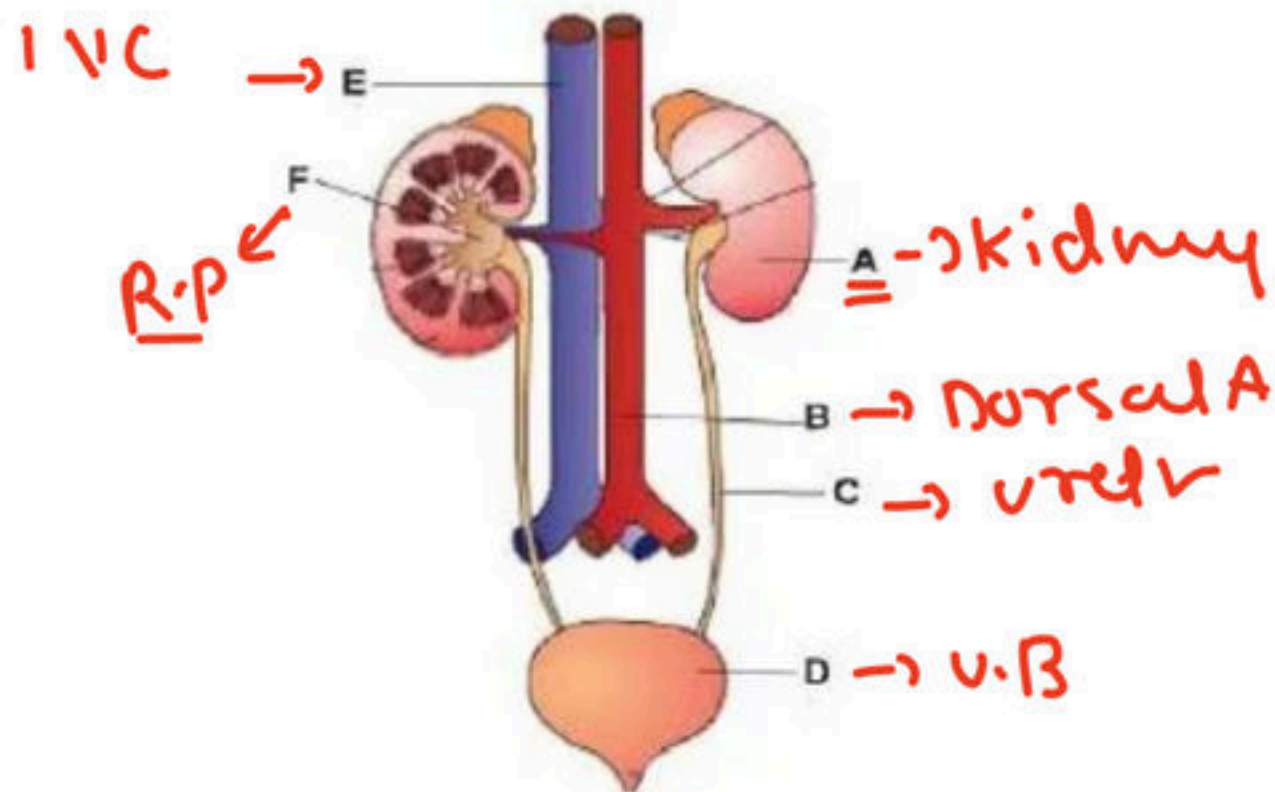
(3) a,d are true

(4) b,c are true





115. In the diagram of excretory system of human beings given below, correctly matched with parts which they represent :



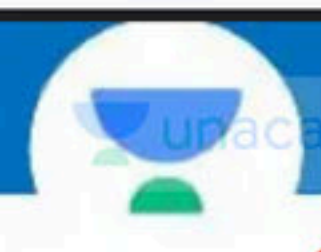
✓ (1) A – Kidney, B – Dorsal aorta, C – Ureter, D – Urinary bladder, E – Inferior venacava, F – Renal pelvis

(2) A – Kidney, B – Inferior venacava, C – Ureter, D – Urinary bladder, E – Dorsal aorta, F – Renal pelvis

(3) A – Kidney, B – Dorsal aorta, C – Urinary bladder, D – Ureter, E – Inferior venacava, F – Renal pelvis

(4) A – Kidney, B – Inferior venacava, C – Urinary bladder, D – Ureter, E – Dorsal aorta, F – Renal pelvis





**116** Given below are two statements :

✓ Statement I : Each half of pectoral girdle consist of a clavicle and a Scapula

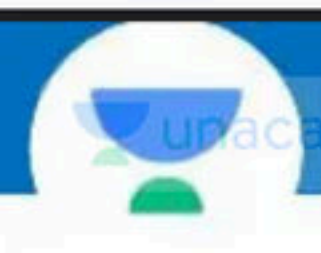
✓ Statement II : Vertebral column, protects the spinal cord, support the head, serves as the points of attachement for the ribs Choose the correct answer from the options given below

(1) Both Statement I and Statement II are incorrect

(2) Statement I is correct but Statement II is incorrect

(3) Statement I is incorrect but Statement II is correct

✓ (4) Both, Statement I and Statement II are correct



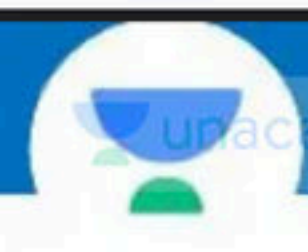
117. Given below are two statements :

- ✗ Statement I : Glenoid cavity is depression to which head of <sup>Humerus</sup> femur articulate ✗
- ✗ Statement II : Parietal bone and temporal bone of skull are jointed by <sup>Fibrous</sup> cartilginous joint

Choose the correct answer from the options given below

- ✓ (1) Both Statement I and Statement II are incorrect
- (2) Statement I is correct but Statement II is incorrect
- (3) Statement I is incorrect but Statement II is correct
- (4) Both, Statement I and Statement II are correct





**118. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R)**

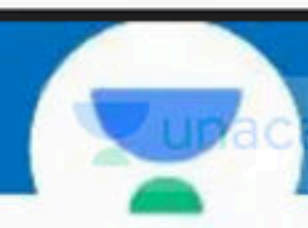
**Assertion (A) :** Gout is inflammation of joints.

**Reason (R) :** Inflammation of joints due to accumulation of uric acid crystals.

**Choose the correct answer from the options given below :**

- (1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- (2) (A) is correct but (R) is not correct
- (3) (A) is not correct but (R) is correct
- ☒ (4) Both (A) and (R) are correct and (R) is the correct explanation of (A)





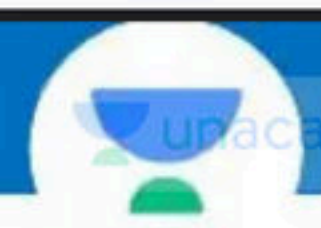
**119. Given below are two statements :**

**Statement I :** Skeletal muscle are closely associated with the skeletal components of the body. ✓

**Statement II :** Based on appearance, cardiac muscles are nonstriated. ✗

**Choose the correct answer from the options given below**

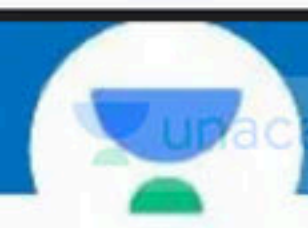
- (1) Both Statement I and Statement II are incorrect
- ✓ (2) Statement I is correct but Statement II is incorrect
- (3) Statement I is incorrect but Statement II is correct
- (4) Both, Statement I and Statement II are correct



120. Number of bones in cranium, face, hyoid and both middle ear are respectively

- (1) 8, 14, 1 and 6
- (2) 8, 14, 1 and 3
- (3) 14, 8, 1 and 3
- (4) 14, 8, 3 and 1





**121.** Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R)

✗ **Assertion (A):** The 8<sup>th</sup>, 9<sup>th</sup> and 10<sup>th</sup> pair of ribs are called floating ribs.

✓ **Reason (R):** Dorsally first seven pairs ribs are attached to thoracic vertebra and ventrally connected to the sternum with the help of hyaline cartilage.

**In the light of the above statements, choose the correct answer from the options given below :**

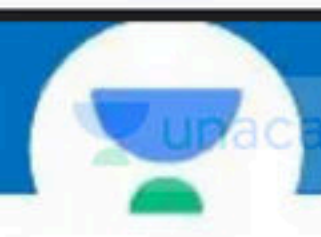
(1) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(2) (A) is true but (R) is false

✓ (3) (A) is false but (R) is true

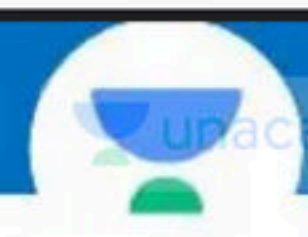
(4) Both (A) and (R) are true and (R) is the correct explanation of (A)





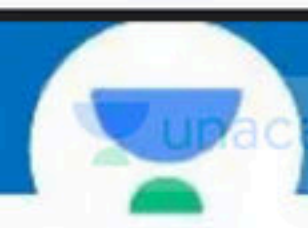
122. Which type of the joint allow limited movement :

- (1) Saddle joint
- ✓ (2) Cartilaginous joints
- (3) Fibrous joints
- (4) Gliding joint



**123** Which is an active process :

- (1) Muscle contraction ✓
- (2) Muscle relaxation ✓
- ✓ (3) Both 1 & 2
- (4) None

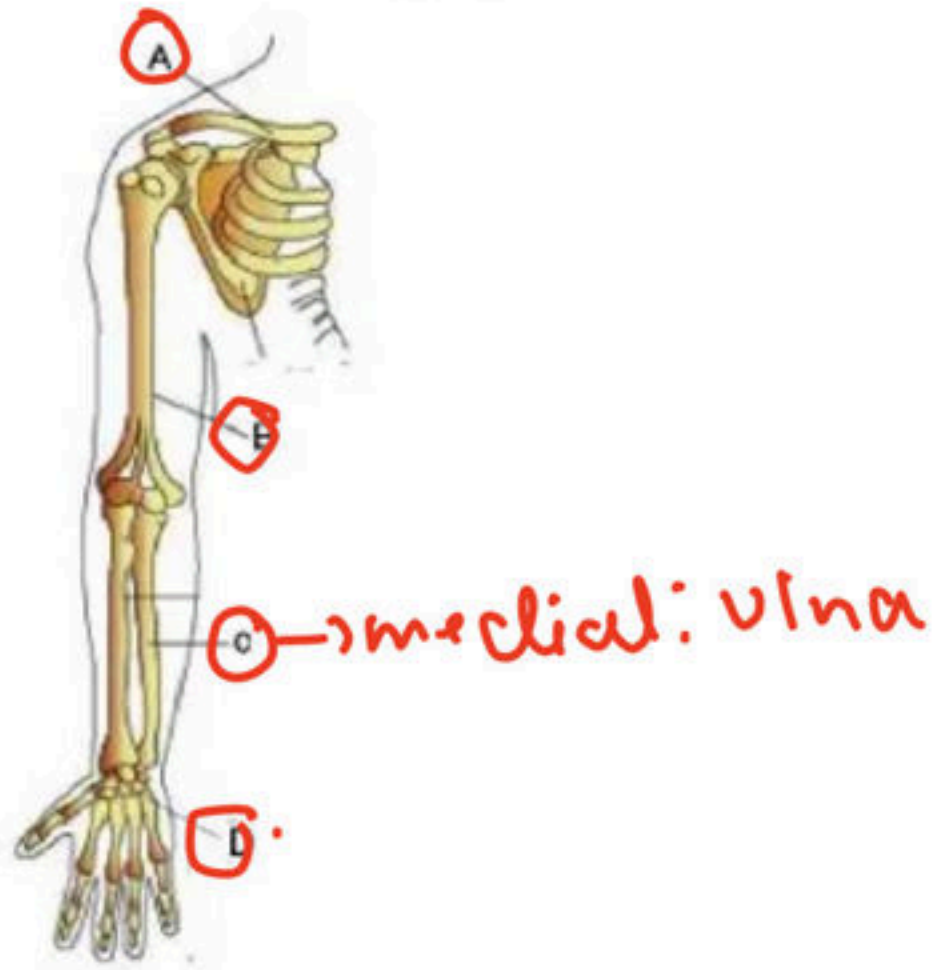


**124. Scapula is a large triangular flat bone situated in the dorsal part of the thorax between**

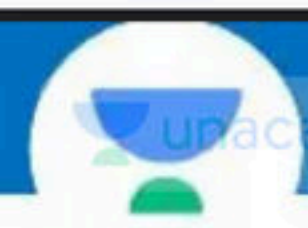
- (1) The second and fifth ribs
- ✓ (2) The second and seventh ribs
- (3) The third and sixth ribs
- (4) The third and eighth ribs



**125. Examine the figure of pectoral girdle and forelimb and identify the parts labelled as A,B, C and D :**

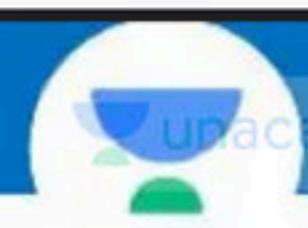


| A            | B       | C      | D       |
|--------------|---------|--------|---------|
| (1) Clavicle | Humerus | Ulna   | Capals  |
| (2) Scapula  | Femur   | Radius | Tarsals |
| (3) Clavicle | Femur   | Radius | Carpals |
| (4) Scapula  | Humerus | Ulna   | Tarsals |



126. Which of the following is store house of calcium ion in striated muscle :

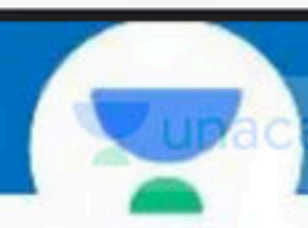
- ✓(1) Endoplasmic reticulum / *Sarcoplasmic Reticulum*
- (2) Mitochondria
- (3) Golgibody
- (4) Sarcolemma



127. Which of the following is/are correct :

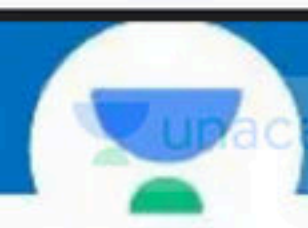
- (1) A complex protein troponin is distributed at regular intervals on tropomyosin
- (2) Progressive degeneration of skeletal muscle mostly due to genetic disorder is called muscular dystrophy ✓
- (3) Inflammation of joints due to accumulation of uric acid crystals is called gout
- ✓ (4) All of these





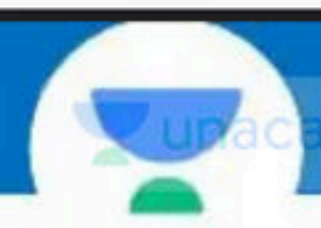
128. Locomotion requires a perfect coordinated activity of ....A....., .....B.....  
and .....C..... system :

| A             | B        | C         |
|---------------|----------|-----------|
| (1) Muscular  | Skeletal | Endocrine |
| ✓(2) Muscular | Skeletal | Neural    |
| (3) Endocrine | Skeletal | Neural    |
| (4) Skeletal  | Muscular | Endocrine |



**129** In following given characteristic of muscles fibres belong to :

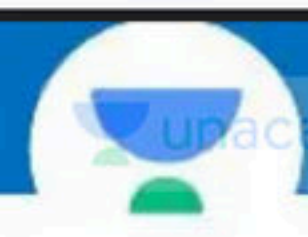
- (a) Aerobic muscles ✓
- (b) Mitochondria are more ✓
- (c) Myoglobin is very high quantity ✓
- (1) Red muscles ✓
- (2) White muscles
- (3) Slow fibre ✓
- ✓ (4) 1 and 3 both



130. Which muscle is though involuntary but structurally similar to skeletal muscle :

- (1) Uterus muscle
- ✓ (2) Heart muscle
- (3) Gastric muscle
- (4) Triceps muscles





131. Which is an correct statement :

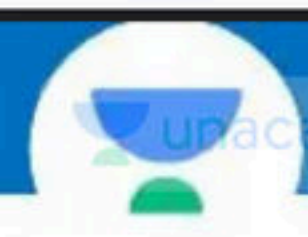
- a- Muscle contraction is a <sup>active</sup> ~~passive~~ process X
- b- Muscles of intestine are involuntary in nature ✓
- c- Fascia is a layer of connective tissue ✓
- d- Muscle bundle is called as ~~sarcomere~~ X

(1) a and b only

✓ (2) b and c only

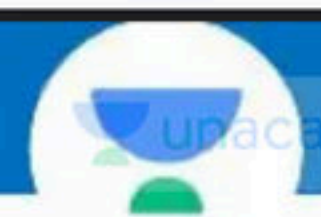
(3) a, b and d

(4) b, c and d



132. Which is a unit of muscle contraction :-

- (1) A band
- ✓ (2) Sarcomere
- (3) Actin filament
- (4) Myosin filament



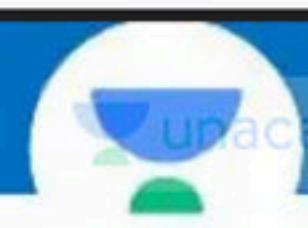
133 Which statement is correct for <sup>"</sup>light band <sup>"</sup> of myofibril:

- ✓ (a) Light band called 'I' band ✓
- (b) Light band called 'A' band ✗
- ✓ (c) Light band is isotropic with polarised light ✓
- (d) <sup>Dark</sup>~~Light~~ band is anisotropic with polarised light

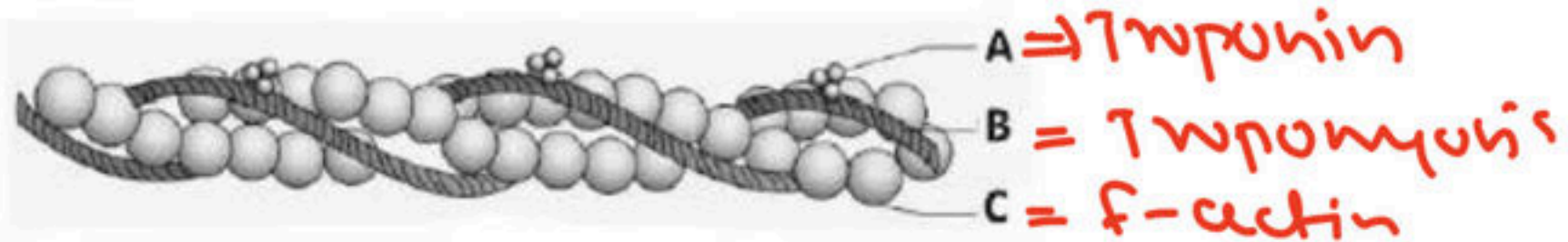
Choose the correct options :

- (1) a and b is correct
- (2) b, c and d is correct
- (3) a and d is correct
- ✓ (4) a and c is correct

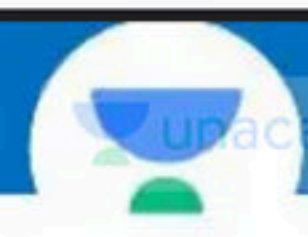




134. Identify A, B and C in an actin filament and choose the correct option :

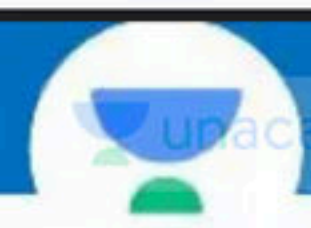


- (1) A–Troponin, B–F actin, C–Tropomyosin
- (2) A– F actin, B–Troponin, C–Tropomyosin
- ✓(3) A–Troponin, B–Tropomyosin, C– F actin
- (4) A–Tropomyosin, B–F actin, C–Troponin



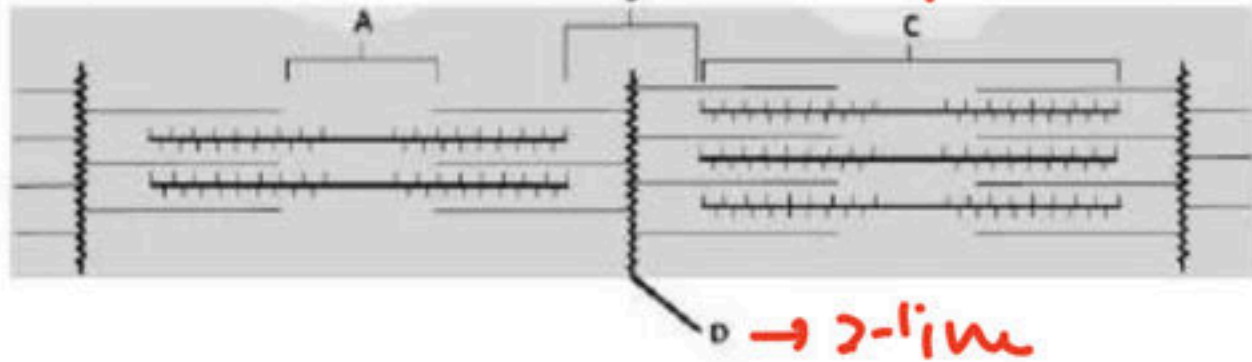
**135. Which statement is incorrect for thick filament muscle fibre :**

- (1) Thick filament is formed by a polymerised protein
- (2) The head with short arm of meromyosin is formed by HMM
- ~~(3)~~ Each thick filament is formed by actin protein ✗
- (4) The tail of meromyosin is formed by LMM



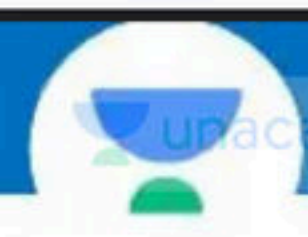
## PART-3 (SECTION-B)

136. Identify A, B, C and D in given two relax sarcomere and choose the correct option :   
 *H2 zone* *I-Band* *A Band*



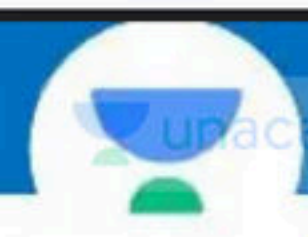
- (1) A-'I' Band, C-'H' Zone
- ☒ (2) B-'I' Band, C-'A' Band
- (3) C-'I' Band, D-'Z' Line
- (4) A-'Z' Line, B-'H' Zone





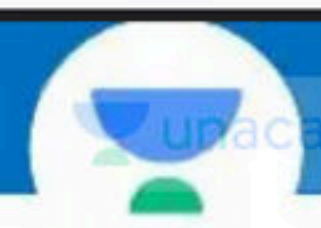
137. Sarcoplasmic reticulum is present in which structure:

- ✓ 1) Muscle
- (2) Neuroglia
- (3) Nervous tissue
- (4) Kidney



**138.** Parallely arranged filaments in the sarcoplasm of skeletal muscle fibre are called as :

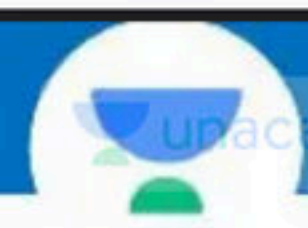
- (1) Myofilaments ✓
- ✓(2) Myofibrils ✓
- ✓(3) Both 1 and 2
- (4) Muscle tone



139. The portion of the myofibril between two successive Z line is :

- (1) Fascia
- ✓ (2) Sarcomere
- (3) H zone
- (4) Both 1 and 3





140. Which is a false statement :

a- Globular head of myosin has ATPase activity ✓

b- F - actin is made up of ~~meromyosins~~ G-actin

c- ~~G-actin~~ F-actin is a polymer of ~~F-actin~~ G-actin

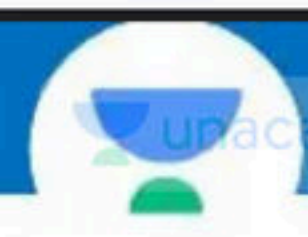
d- L.M.M. is a part of ~~actin~~ myosin

✓ (1) b, c and d

(2) a, b and c

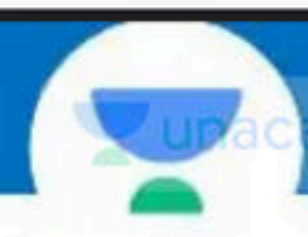
(3) a and c

(4) All



**141. Acetabulum is a part of**

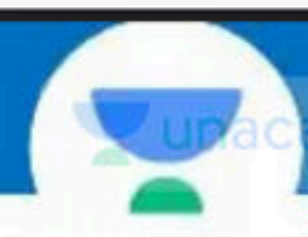
- (1) Pectoral girdle
- ✓ (2) Pelvic girdle
- (3) Femur
- (4) Humerous



142. Which is an incorrect statement :

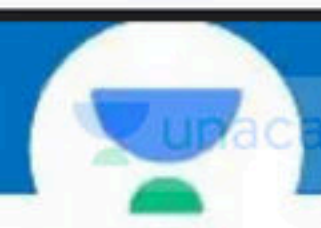
- ✓(1) Dark band is called ~~X~~ band
- (2) Cardiac muscles are striated
- (3) Skeletal muscle fibre is syncytial in nature
- (4) Dark band is called A-band





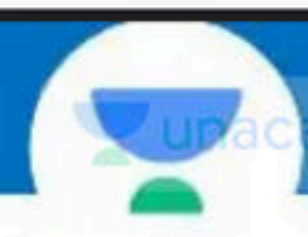
**143.** The globular head of myosin has :

- (1) ATPase enzyme ✓
- (2) Actin binding site ✓
- ✓ (3) Both 1 & 2
- (4) None



**144** Name the thin fibrous proteinaceous membrane which is responsible for the position of thick filament in the middle

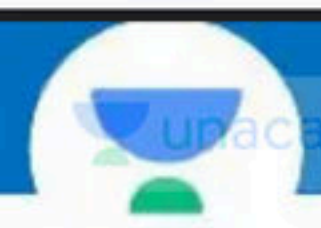
- ✓ (1) M-line ↩
- (2) Z-line
- (3) Actin
- (4) HMM



**145. Which one of the following is correct event during muscle contraction**

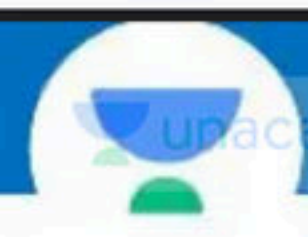
- ✓ (1) Actin slides over myosin towards M-line
- (2) H-zone become wide
- (3) A-bands get reduced
- (4) All of these





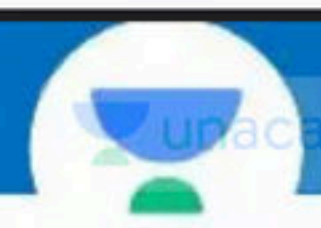
146. In human body, which one of the following is anatomically correct:

- ✓ (1) Floating ribs – 2 pairs
- (2) Collar bones – 3 pairs
- (3) Salivary gland – 1 pair
- (4) Cranial nerves – 10 pairs



**147.** Which one of the following is correct pairing of a body part and the kind of muscle tissue that moves it:

- (1) Heart wall Involuntary ~~un~~striated muscle ✗
- (2) Biceps of upper arm ~~Smooth~~ muscle fibers ✗
- (3) Abdominal Smooth muscle ✓
- (4) Iris Involuntary smooth muscle ✓



**148.** Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R)

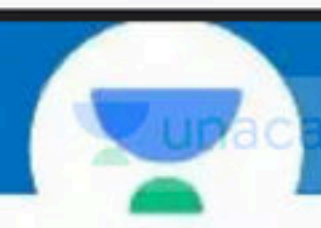
✓ Assertion (A) : Visceral muscles are located in the inner wall of hollow visceral organ in the body.

✗ Reason (R) : Visceral muscles <sup>do not</sup> exhibit straition and are smooth in appearance.

In the light of the above statements, choose the correct answer from the options given below :

- (1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- ✓ (2) (A) is correct but (R) is not correct
- (3) (A) is not correct but (R) is correct
- (4) Both (A) and (R) are correct and (R) is the correct explanation of (A)





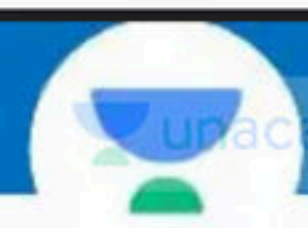
149. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R)

✓ Assertion (A) : Muscles contains a red coloured oxygen storing pigment called myoglobin.

✗ Reason (R) : Myoglobin content is high in all muscle which gives a raddish appearance.  
x  
Red.

In the light of the above statements, choose the correct answer from the options given below :

- (1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- ✓ (2) (A) is correct but (R) is not correct
- (3) (A) is not correct but (R) is correct
- (4) Both (A) and (R) are correct and (R) is the correct explanation of (A)

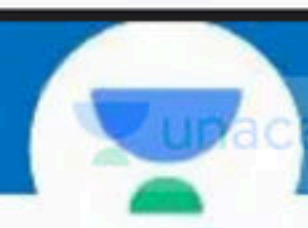


150. Read the following statement and choose correct answer :

- (a) Movement of air into and out of the lungs is carried out by creating a pressure gradient between the lungs and atmosphere ✓
- (b) Inspiration is initiated by contraction of diaphragm which increase the volume of thoracic chamber in dorso-ventral axis ✗
- (c) Contraction of external intercostal muscles lift up the ribs and sternum causing increase in volume antero - posterior axis ✗
- (d) On average a healthy human breathes 12-16 time / minute ✓

- (1) b, d correct and a, c incorrect
- (2) b, c correct and a, d incorrect
- ✓ (3) a, d correct and b, c incorrect
- (4) a, b, c, d correct



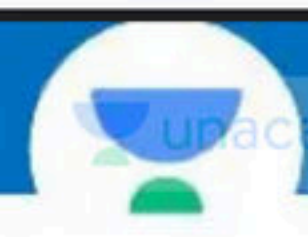
**PART-4 (SECTION-A)**

151. Match the given column I and column II select correct option :

- |                     |                                                           |
|---------------------|-----------------------------------------------------------|
| (A) Bicuspid valve  | (i) Between right atrium and Right ventricle              |
| (B) Semilunar valve | (ii) Between <u>left atrium</u> and <u>Left ventricle</u> |
| (C) Tricupisd valve | (iii) Between <u>right ventricle</u> And pulmonary artery |

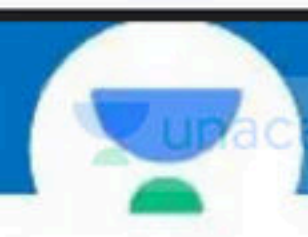
- ✓ (1) A-ii, B-iii, C-i  
(2) A-iii, B-ii, C-i  
(3) A-i, B-iii, C-ii  
(4) A-iii, B-i, C-ii





**152. Which of the following is correct :**

- ✓ (1) Inspiration is an active process
- (2) Inspiration and expiration are passive process
- (3) Expiration is always active process
- (4) Expiration and inspiration are always active process



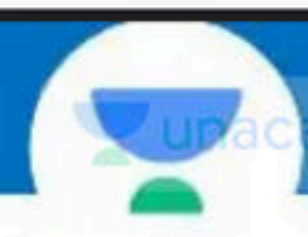
153. The maximum volume of air a person can breath in after forced expiration is : - VC

☒ (1)  $ERV + TV + IRV$

(2)  $ERV + TV$

(3)  $ERV + IRV$

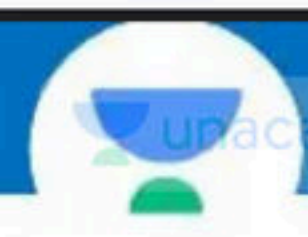
(4)  $IRV + TV$



154. Thoracic chamber is formed dorsally by :

- ✓ (1) Vertebral column →
- (2) Sternum → ventrally
- (3) Diaphragm and ribs → laterally
- (4) All of these

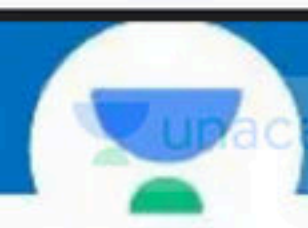




155. Lungs are made up of air-filled sacs, the alveoli. They do not collapse even after forceful expiration, because of :

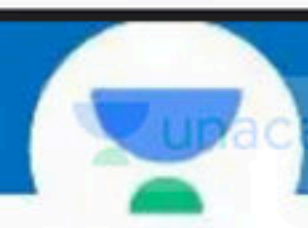
- (1) Inspiratory Reserve Volume
- (2) Tidal Volume
- (3) Expiratory Reserve Volume

✓ (4) R.V



**156.** Volumes of air that will remain in the lungs after normal expiration :

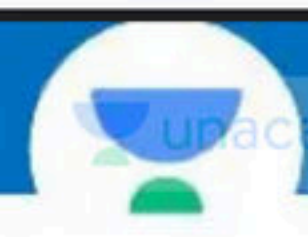
- ✓ (1) E.R.V + R.V = 2300 mL
- (2) E.R.V + R.V = 3600 mL
- (3) E.R.V + T.V = 1600 mL
- (4) E.R.V + T.V = 3500 mL



157. Which of the following factor favour the disociation of  $O_2$  from oxyhaemoglobin :

- (1) Low  $pO_2$ , high  $pCO_2$ , high  $H^+$  conc.
- (2) Low  $pO_2$ , high  $pCO_2$ , low pH
- (3) Low  $pO_2$ , low  $H^+$  conc., high pH
- (4) ☒ Both 1 and 2

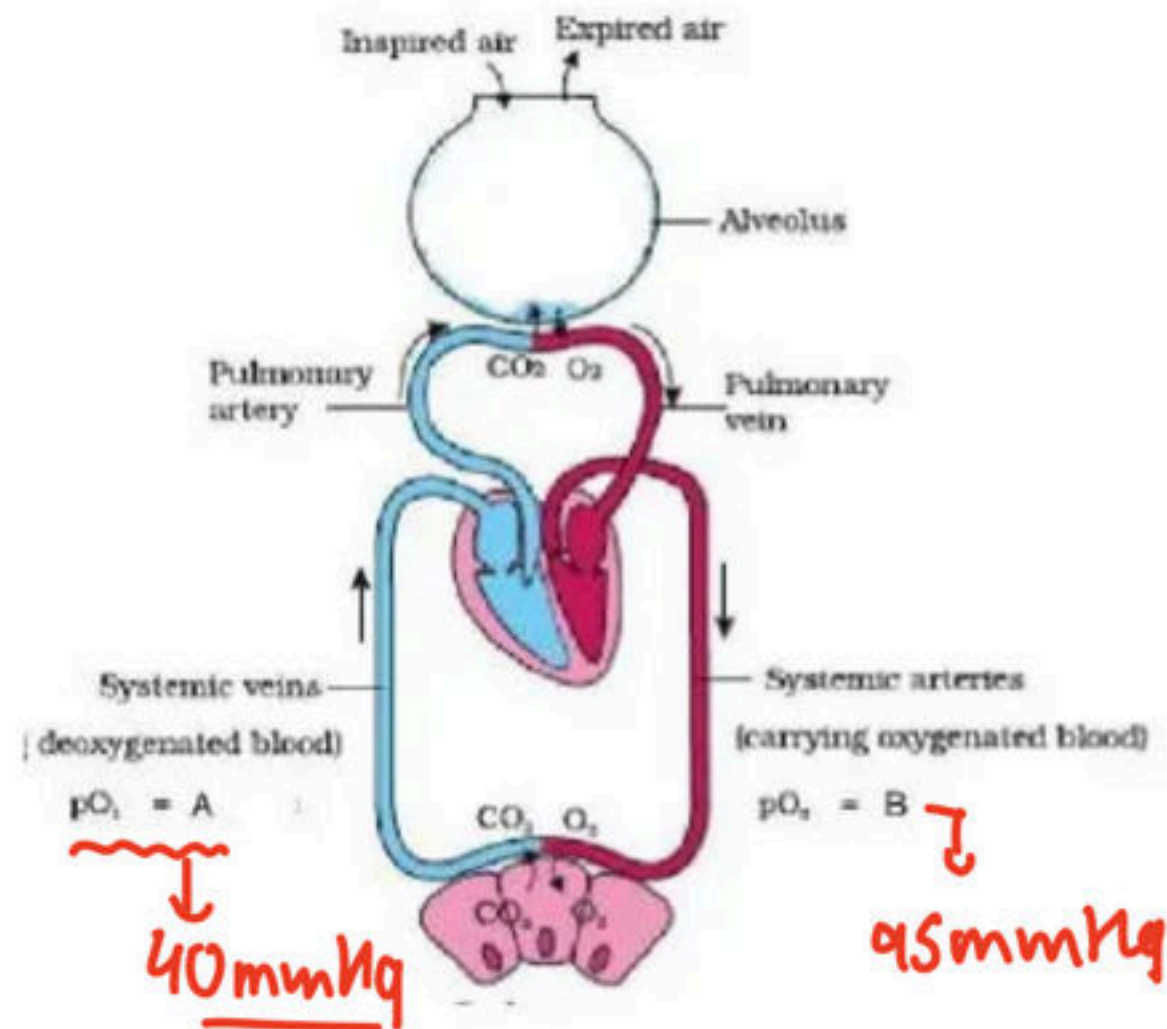




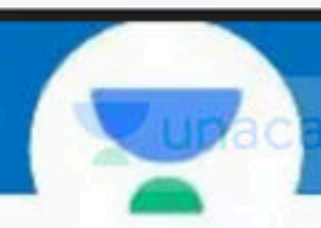
**158. Which of the following factor affect the rate of diffusion of gases :**

- (1) Solubility of gases ✓
- (2) Partial pressure / conc. gradient of diffusing gas ✓
- (3) Thickness of diffusion membrane ✓
- ✓ (4) All of these

159 Read the given figure and select correct option :



- ☒ (1) A – 40 mm Hg, B – 95 mm Hg
- (2) A – 45 mm Hg, B – 40 mm Hg
- (3) A – 40 mm Hg, B – 104 mm Hg
- (4) A – 45 mm Hg, B – 95 mm Hg



160. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

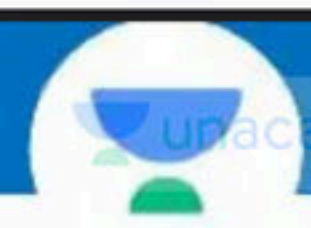
✓ **Assertion (A)** : Pneumotaxic center present in the pons region of the brain.

✗ **Reason (R)** : Pneumotaxic center can moderate the function of respiratory rhythm center.

In the light of the above statements, choose the correct answer from the options given below

- ✓ (1) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (2) (A) is true but (R) is false
- (3) (A) is false but (R) is true
- (4) Both (A) and (R) are true and (R) is the correction explanation of (A)





161. Given below are two statements :

**Statement I :**

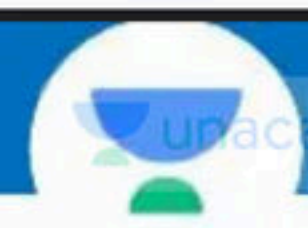
✓ The maximum volume of air a person can breaths in after forced expiration is called vital capacity.

**Statement-II :**

✗ The maximum volume of air a person can breath out after forced inspiration is called expiration capacity. ✗

**Choose the correct answer from the option given below:**

- (1) Both Statement I and Statement II are incorrect
- ✓ (2) Statement I is correct but Statement II is incorrect
- (3) Statement I is incorrect but Statement II is correct
- (4) Both Statement I and Statement II are correct



162. Given below are two statements :

**Statement I :**

Respiratory rhythm center is primarily responsible for regulation of respiration and present in ~~pens~~ pons region of brain.

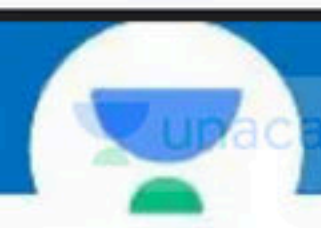
**Statement-II :** *medulla*

A specialised center present in ~~medulla~~ *Pons* region of brain is called pneumotaxic center

**Choose the correct answer from the option given below:**

- ☒ (1) Both Statement I and Statement II are incorrect
- (2) Statement I is correct but Statement II is incorrect
- (3) Statement I is incorrect but Statement II is correct
- (4) Both Statement I and Statement II are correct





163. Match column - I with column - II and select the correct option from the codes given below :

Column - I

A. TV + ERV

B. RV + ERV + TV + IRV

C. ERV + RV

Column - II

(i) Expiratory capacity

(ii) Total lung capacity

(iii) Functional residual capacity

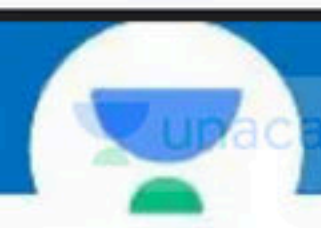
✓ (1) A - (i), B - (ii), C - (iii)

(2) A - (iii), B - (i), C - (ii)

(3) A - (iii), B - (ii), C - (i)

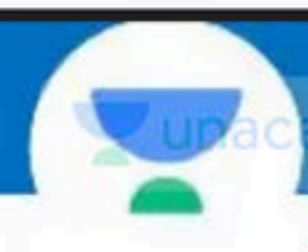
(4) A - (ii), B - (iii), C - (i)





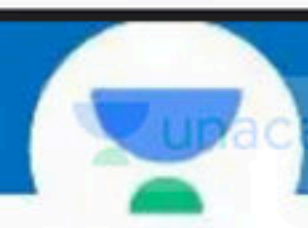
**164. Mark the true statement among the following with reference to normal breathing**

- (1) Inspiration is a passive process where as expiration is active
- ✓ (2) Inspiration is a active process where as expiration is passive
- (3) Inspiration and expiration are active processes
- (4) inspiration and expiration are passive processes



**165** Respiratory process is regulated by certain specialized centres in the brain. One of the following listed centres can reduce the inspiratory duration upon stimulation

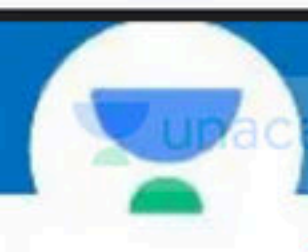
- (1) Medullary inspiratory centre
- ✓ (2) Pneumotaxic centre
- (3) Apneustic centre
- (4) Chemosensitive centre



166. Which of the following match is <sup>" "</sup>wrong regarding transport of gases during respiration :-

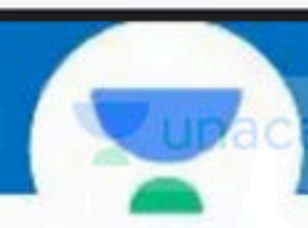
- (1) 97% O<sub>2</sub>, 20-25% CO<sub>2</sub> RBCs
- (2) 3% O<sub>2</sub> dissolved in plasma
- (3) 70% CO<sub>2</sub> bicarbonate
- ✓ (4) ~~3%~~ <sup>7%</sup> CO<sub>2</sub> dissolved in plasma





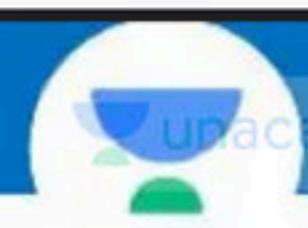
**167.** Which statement is correct for residual volume :

- (1) Volume of air inspired or expired during a normal respiration
- (2) Additional volume of air, a person can inspire by a forcible inspiration
- (3) Additional volume of air, a person can expire by a forcible expiration
- ✓ (4) Volume of air remaining in the lungs even after a forcible expiration



**168. Additional volume of air, a person can inspire by a forcible inspiration is called :**

- (1) Residual Volume
- (2) Expiratory Reserve Volume
- ✓(3) Inspiratory Reserve Volume ✓
- (4) Tidal Volume

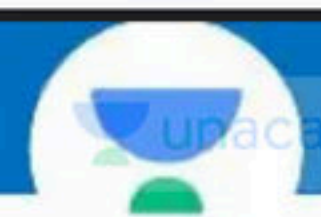


169. Which is a correct statement :-

- (a) Trachea is a straight tube extending upto the mid <sup>thoracic</sup> ~~abdominal~~ cavity ✗
- (b) Pleural fluid is present between two pleural membrane ✓
- (c) Thoracic chamber is dorsally formed by Vertebral column ✓
- (d) We can not directly alter the pulmonary volume ✓

- (1) Only a and b
- (2) Only b and c
- (3) Only a
- ✓ (4) Only b,c,d





170. Which is not a true statement

(a)  $p\text{CO}_2$  of deoxygenated blood is ~~40~~<sup>45 mm Hg</sup> mm Hg

(b)  $p\text{CO}_2$  of alveolar air is 40 mm Hg

(c)  $p\text{O}_2$  of alveolar air is 104 mm Hg

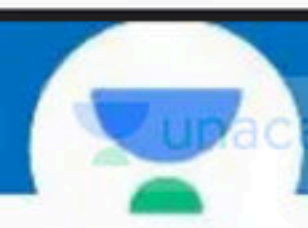
(d)  $p\text{CO}_2$  of deoxygenated blood is ~~95~~<sup>45</sup> mm Hg

(1) Only a and b

(2) Only b and c

(3) Only d

☒ (4) Only a and d



171. Which is an correct statement :-

(a) Spirometer has clinical significance ✓

(b) A healthy man can inspire or expire approx-  
6000 to 8000 ml of air per second ✗

(c) Expiratory reserve volume is 1000 ml ✓

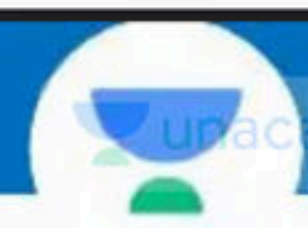
(d) Normally, expiration is a passive process ✓

(1) Only a and b

(2) Only b and c

✓ (3) Only a, c, d

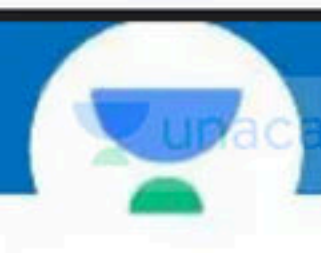
(4) a, b, c, d



172. Which statement is incorrect :

- (1) Lower invertebrates like sponges, coelenterates, flatworms, etc., exchange  $O_2$  with  $CO_2$  by simple diffusion over their entire body surface.✓
- (2) Trachea is a straight tube extending up to the midthoracic cavity, which divides at the level of 5<sup>th</sup> thoracic vertebra into a right and left primary bronchi.✓
- (3)  $pO_2$  is a major factor which could affect binding of  $CO_2$  with haemoglobin.✓
- ✓ (4) None of these





173. Given below are two statements :

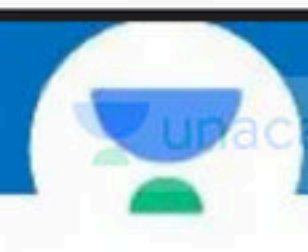
**Statement I :**

Every 100 ml of deoxygenated blood delivers approximately ~~8~~<sup>4</sup> ml of CO<sub>2</sub> to the alveoli. ✗

**Statement-II :**

✓ CO<sub>2</sub> is carried by heamoglobin as carbaminoheamoglobin. **Choose the correct answer from the option given below:**

- (1) Both Statement I and Statement II are incorrect
- (2) Statement I is correct but Statement II is incorrect
- ✓ (3) Statement I is incorrect but Statement II is correct
- (4) Both Statement I and Statement II are correct



174. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

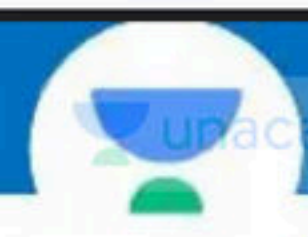
✓ **Assertion (A)** : Solubility of CO<sub>2</sub> is 20–25 time higher than that of O<sub>2</sub>. ↕

✓ **Reason (R)** : The amount of CO<sub>2</sub> that can diffuse through the diffusion membrane per unit difference in partial pressure is much higher compared to that of O<sub>2</sub>.

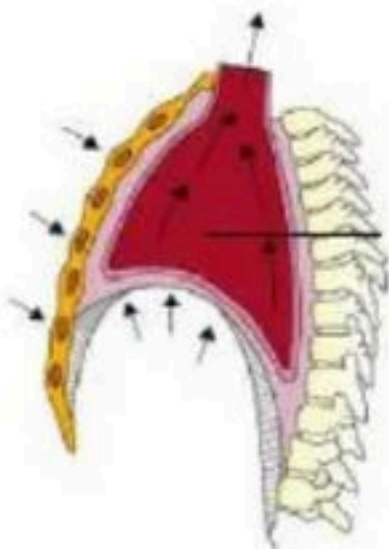
**In the light of the above statements, choose the correct answer from the options given below**

- ✓ (1) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (2) (A) is true but (R) is false
- (3) (A) is false but (R) is true
- (4) Both (A) and (R) are true and (R) is the correction explanation of (A)





175. Following given figure represents :



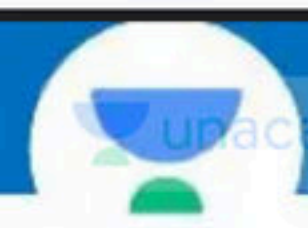
(1) Expiration ✓✓

(2) Diaphragm relaxed ✓✓

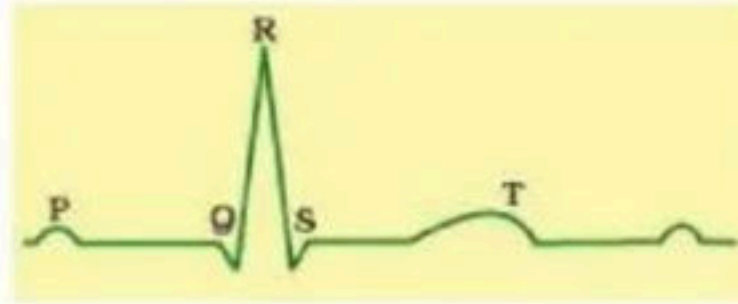
(3) Inspiration

✓✓ (4) 1 and 2 both

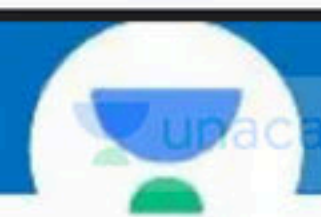




176. In given diagrammatic representation T represent:



- (1) Depolarisation of Atria
- (2) Beginning of the systole
- ✓ (3) Repolarisation of ventricles
- (4) End of diastole



177. Match the types of W.B.C. with their percentage in W.B.C.

Types of W.B.C                      % in W.B.C

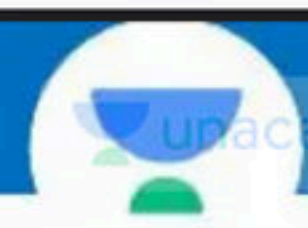
- |                |                   |
|----------------|-------------------|
| a. Basophils   | i. 2 - 3          |
| b. Neutrophils | ii. 20 - 25       |
| c. Eosinophils | iii. 6 - 8        |
| d. Monocytes   | iv. 60 - 65       |
| e. Lymphocytes | v. <u>0.5 - 1</u> |
- 

(1) a - v, b - iv, c - iii, d - i, e - ii

~~(2) a - v, b - iv, c - i, d - iii, e - ii~~

(3) a - i, b - iv, c - v, d - iii, e - ii

(4) a - ii, b - iv, c - i, d - iii, e - v



178. Read the following events of blood clotting and choose correct sequence in options :

- (i) Conversion of fibrinogen to fibrin → ③
- (ii) Formation of clot - ④
- (iii) Conversion of prothrombin to thrombin → ②
- (iv) Thromboplastin formation → ①

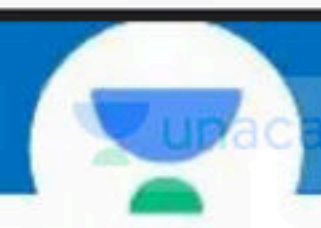
~~(1)~~ iv - iii - i - ii

(2) iii - iv - i - ii

(3) i - iv - iii - ii

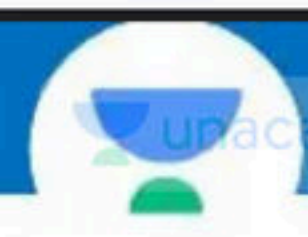
(4) i - iii - iv - ii





**179. Blood pressure in mammalian aorta is maximum during :**

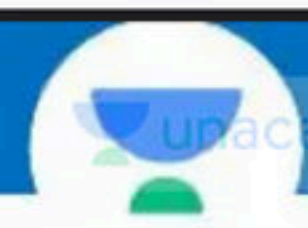
- (1) Diastole of the right atrium
- (2) Diastole of left atrium
- ☒ (3) Systole of left ventricular
- (4) Joint diastole



**180. Second heart sound dub is associated with :**

- ✓ (1) Closure of the semilunar valves ↩
- (2) Closure of cuspid valves
- (3) Opening of semilunar valves
- (4) Opening of cuspid

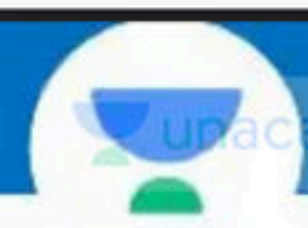




**181. Which of the following statement is incorrect :**

- (1) A person of O blood group has anti A and anti B antibodies in blood plasma
- (2) ABO Blood group is based on the basis of presence or absence of antigen in blood plasma
- (3) Person of AB blood group is universal recipient
- (4) Person of O blood group is universal donors





**182. Match the column I and column II**

**Column I**

**Column II**

**a. Basophils**

**i. Inflammation**

**b. Neutrophils**

**ii. Blood clotting**

**c. Plasma cells**

**iii. Phagocytosis**

**d. Platelets**

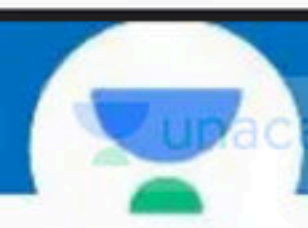
**iv. Antibodies**

(1) a - i, b - iii, c - iv, d - ii

(2) a - i, b - iii, c - ii, d - iv

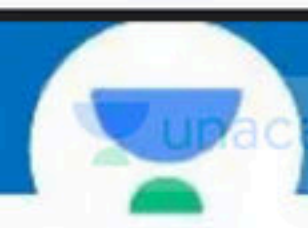
(3) a - iii, b - ii, c - iv, d - i

(4) a - iii, b - i, c - iv, d - ii



**183. Which of the following is incorrect :**

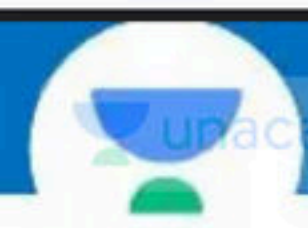
- (1) Deposition of calcium, cholesterol and fibrous tissue in coronary artery called antherosclerosis
- (2) Angina commonly occur during middle age
- (3) 120/80 is normal blood pressure
- (4) When heart muscles suddenly damaged by an inadequate blood supply is called heart failure



**184. *Erythroblastosis foetalis* can be avoided by administering ..... A ..... to the mother immediately after the delivery of first child :**

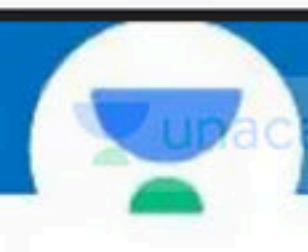
- (1) A - anti Rh antibodies
- (2) A - Rh antibodies
- (3) A - Anti Rh antigen
- (4) A - Rh antigen





**185. The central part of thick filament which is not overlapped by thin filaments is called as :**

- (1) Z zone
- (2) H-zone
- (3) I-zone
- (4) F-zone



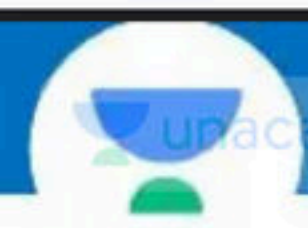
## PART-4 (SECTION-B)

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**186. Choose the incorrect statement :**

- a- Dark band is called Isotropic-band**
- b- Cardiac muscles are striated and voluntary**
- c- Z line bisects I-band**

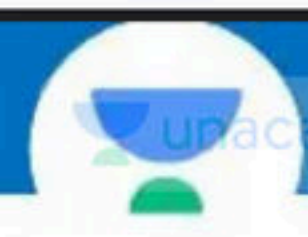
- (1) a and b only
- (2) b and c only
- (3) a and c only
- (4) a, b and c



**187. ATPase enzyme, essential for muscle contraction is found in :**

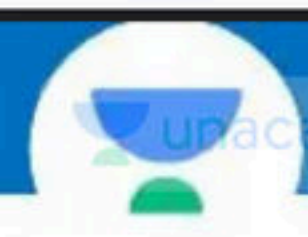
- (1) Actin
- (2) Myosin
- (3) Troponin
- (4) Tropomyosin





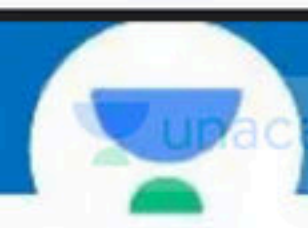
**188. In the resting state a subunit of troponin masks the active binding sites for myosin on :**

- (1) Actin filaments
- (2) Meromyosin filament
- (3) M–line
- (4) H–zone



**189. During contraction of skeletal muscle,  $\text{Ca}^{2+}$  bind to :**

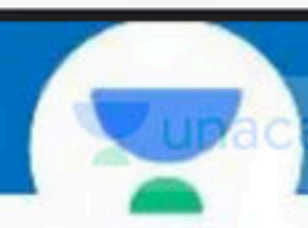
- (1) Myosin
- (2) Troponin
- (3) Tropomyosin
- (4) Myosin ATPase



**190. Which statement is correct for muscle contraction:**

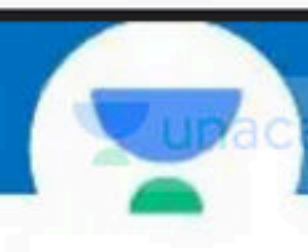
- (1) Length of H-zone decreases
- (2) Length of A-band decreases
- (3) Length of two Z-lines increases
- (4) Length of I-band remains constant





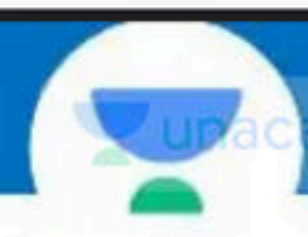
**191. On violent exercise such as fast running causes muscular fatigue and cramps due to :**

- (1) Uric acid accumulation
- (2) Calcium oxalate accumulation
- (3) Shortening of tendons
- (4) Lactic acid accumulation



**192. Pelvic girdle consists of :**

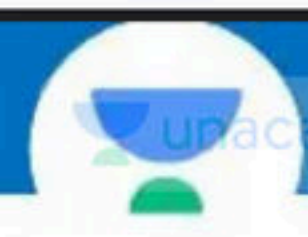
- (1) Ilium only
- (2) Ilium and ischium only
- (3) Ischium and pubis only
- (4) Ilium, ischium and pubis



**193. Which of the following is a synovial joint :**

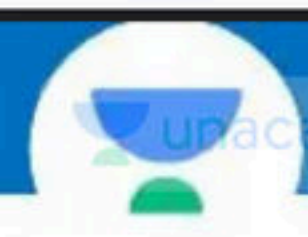
- (1) Ball and socket joint
- (2) Hinge joint
- (3) Fibrous joint
- (4) Both 1 & 2





**194. Wild contractions of muscles due to low  $\text{Ca}^{2+}$  in body fluid is called as :**

- (1) Arthritis
- (2) Tetany
- (3) Osteoporosis
- (4) Gout



**195. Which statement is incorrect for limb :**

- (1) Tarsals are 7 in number
- (2) Metatarsals are 5 in number
- (3) Carpals are 8 in number
- (4) Humerus is the longest bone



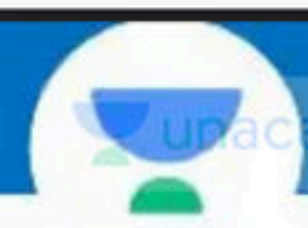
**196. Read the following statement :**

- (a) Muscle is a specialised tissue of endodermal origin**
- (b) Muscle is a specialised tissue of mesodermal origin**
- (c) About 70-80 per cent of the body weight of a human adult is contributed by muscles**
- (d) Skeletal muscles have a striped appearance**

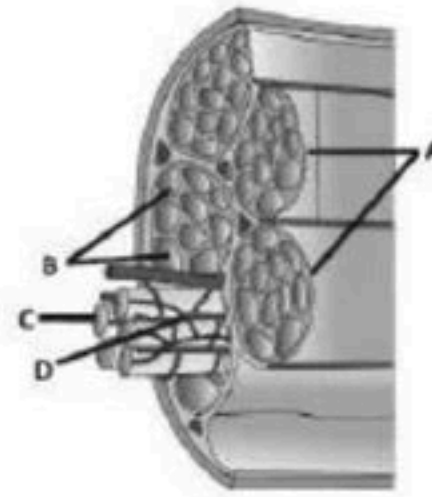
**Choose the correct options :**

- |                               |                                  |
|-------------------------------|----------------------------------|
| <b>(1) a and b is correct</b> | <b>(2) b, c and d is correct</b> |
| <b>(3) b and d is correct</b> | <b>(4) a and c is correct</b>    |

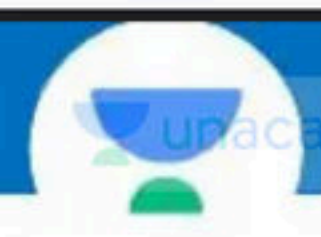




197. Identify A, B, C and D in diagram of cross section of muscles & choose correct option :

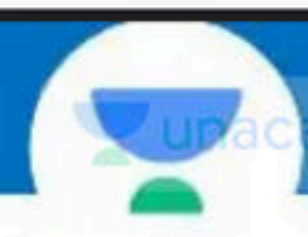


- (1) A–Blood capillary, C–Sarcolemma
- (2) B–Fascicle, C–Sarcolemma
- (3) C–Blood capillary, D–Sarcolemma
- (4) A–Fascicle, B–Muscle fibre



**198. Zygomorphic symmetry is present mostly in:**

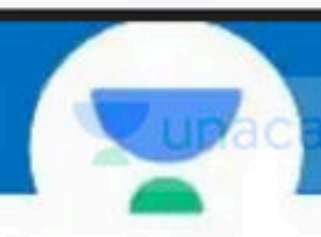
- (1) Malvaceae
- (2) Gramineae
- (3) Crcuciferae
- (4) Compositae



**199. Superior ovary is/are present in :**

- (1) Malvaceae
- (2) Leguminosae
- (3) Gramineae
- (4) All of these





**200. In okra, cotton, mustard, cauliflower, sunflower marigold, maize. how many belong to family malvaceae:**

- (1) Two
- (2) Three
- (3) Five
- (4) Six